Item-dropping to improve in-sample Cronbach's alpha can harm true reliability: A Monte Carlo simulation

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Abstract

Cronbach's alpha is the most commonly reported metric of reliability in psychological measurement, and statistical software encourages a common post hoc revision: dropping the item whose removal most improves in-sample alpha. We examined what this practice does to a scale's true reliability in a Monte Carlo simulation varying items per scale (5 to 40) and sample size (20 to 1,000), with item pools generated from a loading distribution fitted to 3,783 factor loadings from real psychological scales. Two strategies, always dropping the nominated item or dropping only when in-sample alpha improves, were evaluated against the population alpha and the true reliability of the items actually retained. The bias in the reported alpha itself was modest, overstating true reliability only in short scales at small samples and understating it elsewhere. The decision was another matter: at the smallest scale and sample size, dropping reduced true reliability in 46.7% of the replications in which the conditional rule acted, despite acting only when alpha appeared to improve, and almost the entire apparent gain was selection optimism rather than real improvement. Stricter decision rules helped only in large samples: with N of 200 or more, acting only on apparent gains above .02 kept the probability of harming the scale at or below 7%, whereas at N = 20 no threshold examined made the decision safe. Because this risk depends on scale length and sample size, given loadings typical of trait scales, it is knowable before data are collected, whereas nothing inside the alpha-if-item-removed table distinguishes a genuinely weak item from a sampling artefact.

1. Introduction

Cronbach's α provides a lower bound of the internal consistency of a multi-item scale under the assumption of tau equivalence (Cronbach, 1951). It is the most widely reported metric of structural validity in psychology (Flake et al., 2017) and also sees wide use in related fields, despite longstanding critique of its use and interpretation (Cortina, 1993; Sijtsma, 2009). Although conventions for acceptable reliability are less universal for α than for other metrics (e.g., the $p < .05$ convention for statistical significance), there

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is commonly pressure to meet a threshold of reliability. Indeed, Hussey et al. (2025) extracted more than 180,000 reported α values from three very large corpora, covering the general psychology literature, the industrial and organizational psychology literature, and the American Psychological Association's PsycTests database, and found robust excesses of values sitting just above the .70 rule-of-thumb threshold that could not be explained by justifiable measurement practices. They give this the name α-hacking, defined as altering a measure ad hoc after seeing the results, with the goal or effect of raising the in-sample α, thereby conditioning the analytic choices on the results and overfitting to the data at hand. Because their approach can detect only those distortions that land an α at one of the thresholds, and not the presumably far more common ones that land it anywhere else, they describe what they observe as an extreme lower bound on the true prevalence.

There have been calls for an increased focus on measurement when considering the validity of study conclusions (Flake & Fried, 2020). Flake and Fried emphasise the need for transparency in order to assess questionable measurement practices, that is, decisions researchers make that raise doubts about the validity of the measures in a study and ultimately about its conclusions. Item dropping on the basis of Cronbach's α is a particularly clear instance, because it is both ubiquitous and actively prompted by the software researchers use, and because the same sample supplies both the decision and the number subsequently reported.

Improving psychological measurement is desirable, and much work is done on scale development and refinement to precisely measure constructs of interest (e.g., Clark & Watson, 1995, 2019; Jebb et al., 2021; Rodriguez et al., 2016). In the pursuit of more reliable scales, it is plausible that some modifications occurring after data collection (e.g., item dropping, reverse scoring) may be influenced by the pressure to report a sample α that meets a common threshold. Such changes would reduce the validity of the scale's interpretation, since it is unclear that in-sample gains in reliability observed in a single sample would generalise to out-of-sample contexts. The flexible application of scale modification, such as through item dropping, can represent a questionable measurement practice (Flake & Fried, 2020; Hussey et al., 2025; Hussey & Hughes, 2020). It can be difficult to distinguish scientifically desirable cases of item dropping from undesirable ones. Cronbach's α is partially a function of scale length, so shortening a scale exerts downward pressure on α. However, items are presumably dropped in studies when they worsen the full-scale α, which exerts upward pressure. The summative impact of these two opposing pressures on α has not been investigated.

Selection of items from an item pool is a key step in the development of a scale (Clark & Watson, 2019), and can be an important part of ongoing scale refinement. However, item dropping can also occur in more ad hoc contexts. It can be used in ways that cannot aid cumulative understanding of scale reliability, for example when it is not reported (Heggestad et al., 2019). Elson et al. (2023) make this point directly: researchers frequently drop items they argue to be poorly performing in order to boost a scale's reliability coefficient, but without an out-of-sample norm for the modified measure, or a replication showing that the same drop consistently improves measurement, doing so improves neither the present nor the future use of the measure, and the modification is often not fully reported in any case. Item dropping can serve to improve α in a given sample, for example to meet a cut-off value for acceptable reliability, or as an experimenter degree of freedom that may influence the results of a subsequent hypothesis test using scores on the scale. Both represent instances of overfitting, or of conditioning analyses on their results, which reduces the replicability and validity of claims. Such overfitting is also unlikely to be caught. Psychological measures are overwhelmingly used once and then abandoned: 43% appear only a single time in the APA PsycInfo record of tests used in studies (Elson et al., 2023), and 79% are not reused more than twice (Anvari, Alsalti, Oehler, Hussey, et al., 2025; Anvari, Alsalti, Oehler, Marion, et al., 2025). For most measures there is therefore never a second sample in which an apparent in-sample gain could fail to replicate, and a researcher may be overfitting considerably more than they realise. Distortions in the α values reported in articles serve to undermine the credibility of the broader literature.

The downstream cost of this particular practice is already established. Stefan & Schönbrodt (2023) show by simulation that item dropping based on Cronbach's α inflates the false positive rate of subsequent hypothesis tests from the nominal 5% to as high as 30%. What that result does not tell us is why, or how often. An inflated false positive rate downstream is consistent with several quite different upstream stories: that the reported α is systematically too high, that the retained scale is genuinely worse than the one it replaced, or simply that the analyst has acquired an extra degree of freedom. These have different implications for practice, and distinguishing between them requires measuring what item dropping does to the measurement, not only to the test that follows it.

The measurement side has one main precedent. Kopalle & Lehmann (1997) examined it with simulated data, using an alpha-if-item-removed selection method that drops whichever single item maximises sample α, and showed that selecting on the same sample used to report the resulting α capitalises on chance. Their evaluation established the phenomenon, but the precision of their estimates is limited by the use of only 50 iterations per condition, their loading distributions and pool sizes were chosen arbitrarily rather than estimated from real scales, and the reported α was evaluated against the population α of the retained items rather than against their true reliability, which is the quantity α is reported as estimating. We therefore do not know the relative trade-off between the improvement in reliability from dropping bad items and the worsening caused by shortening the scale: the marginal cost of retaining a mediocre item may well be smaller than the benefit of the scale being longer.

This article informs this discussion by asking whether item dropping, on average and in the individual case, improves or worsens estimation of the true reliability of a scale. We focus on one item-selection mechanism, maximising alpha-if-item-removed, because of its ubiquitous availability in statistical software: SPSS, JASP, jamovi, and the R packages *psych* and *performance* all present it (Lüdecke et al., 2021; Revelle, 2024). The user is shown the α observed in the study sample alongside a table of what α would be if each scale item were excluded in turn. Hussey et al. (2025) argue that presenting this information may itself act as a cue that raises the probability of a researcher dropping an item post hoc, without prompting any deeper engagement with the relative performance of the items or with the appropriateness of removing one. The software highlights an action that would improve the reported α, and gives no indication of the bias that action could introduce.

The same authors pose the question this simulation is built to answer: without replication in a new sample, how can researchers know whether a given item was genuinely performing poorly and should be dropped, or whether they are simply conditioning their analysis on their own results? In real data that question is unanswerable, because the truth is not observed. In a simulation it is answerable directly, because the population loadings are known, so the two cases can be distinguished within every simulated study and the rate at which they are confused can be estimated. By extending the simulation approach of Kopalle and Lehmann with an empirically grounded data-generating process, substantially more replications, and population targets that separate the reported number from the underlying decision, the current work quantifies the effects of item dropping on both what is reported and what is measured.

2. Method

We simulated the impact of item dropping using a data-generating process with parameters estimated from a large corpus of empirical self-report data from a wide range of psychological scales, so that the simulated scales resemble the reliability properties of real trait scales rather than an arbitrary parametric grid.

The ADEMP framework (Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures) is a structured framework for designing and reporting simulation studies for maximum interpretability and interoperability. It was originally proposed for simulation studies in biostatistics (Morris et al., 2019) and has since been adapted and promoted for psychology, including a preregistration template (Siepe et al., 2024). We follow this structure throughout the remainder of this section, so that the design can be reused independently of the narrative text. The full research compendium, containing the complete ADEMP specification, the validation of the fast reliability functions against `psych::alpha()`, and per-condition tables for every quantity reported here, is available in the project repository (see *Data and code availability*).

2.1. Aims

The statistical task is estimation. The estimator under study is the sample Cronbach's α of the retained items, that is, the number the analyst actually reports, evaluated against two population targets defined for the specific items retained. The aims are:

1. Calibration and the structural gap (full pool, no dropping). To estimate the bias of sample α as an estimator of the true population α of the full item pool, which serves as a check that the simulation pipeline behaves as expected, and to quantify the baseline gap between population α and true reliability that exists before any item selection.
2. Bias in the reported reliability of the retained items. For each strategy, to estimate how much the sample α of the retained items over- or under-states (a) the true population α of those same items and (b) their true reliability.
3. What dropping actually does to the scale. To estimate the expected change from full pool to retained items on three scales (the observed sample α, the population α, and the true reliability), the optimism gap between the first two, and the proportion of replications in which dropping actually reduced true reliability.

4. How much a single study can gain or be misled. Aims 1 to 3 are long-run expectations and are silent on the tail. To address it we estimate the spread of the per-replication quantities, summarised as 95% prediction intervals, alongside every mean reported under Aims 2 and 3.

2.2. Data

For the empirically fitted data-generating process we used *granary*, a very large aggregation of other open datasets of granular item-level psychological traits and attitudes (Hussey, 2026). Single-item measures do not yield a factor loading and were excluded. For each pairing of scale and study source, participants who completed fewer than 80% of the items from that measure were excluded prior to any model fitting, using a threshold applied uniformly across datasets.

2.2.1. Fitted parametric distribution of loadings

The pooled distribution comprised 3,783 standardised single-factor loadings across 206 (scale, source) units, contributed by 196,769 distinct participants. Each loading comes from a separate single-factor confirmatory factor analysis fitted to one (scale, source) dataset rather than from a single pooled model, and because a participant may complete several scales, those 206 analyses together rest on 252,478 scale-level observations. The number of participants per unit is large on average but highly variable and strongly right-skewed: mean 1,226 (SD 2,564), median 468, interquartile range 255 to 1,027, minimum 99, maximum 19,330. Given that skew, the minimum and maximum are reported alongside the mean and standard deviation because the latter pair is on its own misleading here: the standard deviation exceeds twice the mean, 86% of units fall below the mean rather than the mean sitting near their centre, and the largest unit is nearly 200 times the smallest. The median is the more representative summary of a typical contributing scale. Scale length was similarly skewed, with a mean of 18.4 items (SD 15.0), median 13, minimum 3 and maximum 90. Loadings enter the pool unweighted, so each item counts once regardless of the sample size of the scale it came from, which is the appropriate target for a data-generating process that draws items rather than scales.

The pooled distribution was best fit, among six candidate distributions compared by AIC, BIC, and Kolmogorov-Smirnov distance, by a scaled Beta distribution obtained by rescaling the loading onto $[0, 1]$ before fitting:

$$\lambda_i \sim 2 \times \mathrm{Beta}(11.77,\, 3.84) - 1$$

The fitted shape parameters were $\text{shape}_1 \approx 11.77$ and $\text{shape}_2 \approx 3.84$, giving a mean loading of approximately 0.51 (mean inter-item correlation ≈ 0.26) and a left-skewed distribution with an approximately 1.7% negative-loading tail. These shape parameters are estimated in a separate file (Hussey, 2026) and enter this simulation only as fixed constants. The loading distribution is not a factor of this design and is not manipulated. Two of its properties are nonetheless important for the estimands: loadings are unequal, which is what makes population α and true reliability distinct targets, and the small negative tail means some generated pools contain a genuinely α-harming item, so conditional dropping is not always chasing noise.

2.2.2. Distribution of scale lengths

The number of items per scale was recorded for each of the 206 (scale, source) units. This distribution was right-skewed, ranging from 3 to 90 items. Rather than fixing the simulated item-pool size at one value, we took the 10th, 25th, 50th, 75th, and 90th percentiles of this empirical distribution (5, 9, 13, 21, and 39 items respectively, rounded to 5, 10, 15, 20, 40 for even spacing) as the item-pool sizes to simulate. The range of scale lengths studied therefore reflects how long real trait scales tend to be in published research, from short screening instruments to long, multidimensional batteries.

2.3. Data-generating process

Item-level data for one simulated scale were generated per replication from a standardised single-factor model using `lavaan::simulateData()` (Rosseel, 2012). The common factor is standard normal, item i loads λ_i on it, and its unique variance is $\psi_i = 1 - \lambda_i^2$, so that every item has unit total variance and $\mathrm{Cov}(X_i, X_j) = \lambda_i \lambda_j$ for $i \neq j$. Because the population loadings are drawn by the simulation rather than estimated from data, the true population Cronbach's α and the true reliability of the unit-weighted sum score are available in closed form for any subset of items, with no need for a separate large-N reference sample.

2.3.1. Design factors

Two factors were manipulated, crossed fully factorially:

- Item-pool size (k): five levels (5, 10, 15, 20, 40 items), taken from the empirical percentiles described above.
- Sample size (N): five levels (20, 30, 50, 200, 1000), chosen to reflect a range of plausible study sizes in psychological research.

This gives $5 \times 5 = 25$ conditions. Each condition was replicated 10,000 times, for a total of 250,000 independently generated datasets. One dataset is generated per replication and both strategies are applied to it. Because both strategies remove at most one item, there is no "number of items retained" factor: how many items the conditional strategy retains is an outcome of the rule, not a manipulated input. There is no exclusion rule, since Cronbach's α is a closed-form function of the sample covariance matrix and does not require a full-rank correlation matrix, so cells with $N < k$ are retained.

2.4. Item-dropping methods

Both strategies nominate the same candidate item from the sample data,

$$J^* = \arg\max_j \hat{\alpha}(-j)$$

where $\hat{\alpha}(-j)$ is the sample α of the pool with item j removed, and differ only in whether acting on that nomination is gated on an observed improvement:

- Forced drop. Remove item J^* regardless of whether $\hat{\alpha}(-J^*)$ actually exceeds the full-scale $\hat{\alpha}(F)$. The retained set is therefore: $S_{\text{forced}} = F \setminus \{J^*\}$ always.
- Conditional drop. Remove item J^* only if it improves in-sample alpha ($\hat{\alpha}(-J^*) > \hat{\alpha}(F)$), otherwise remove nothing. Therefore: $S_{\text{cond}} = F \setminus \{J^*\}$ or F.

Because the strategies share the nomination rule, comparisons between them are comparisons of the gating rule with the nomination rule held fixed. This is an intended design feature rather than a confound, and it is why the alpha-if-removed vector is computed once per replicate in the code.

2.5. Validation of the implementation

Sample α and the k leave-one-out alphas are computed directly from a single sample covariance matrix rather than by calling `psych::alpha()` once per item, which is what makes a design of this size tractable: the direct computation avoids building the large object of item statistics that `psych::alpha()` returns and that this simulation never uses, reducing the cost of a full run from hours to minutes. Substituting a hand-written implementation for the field-standard one is only defensible if the two are shown to agree, so a separate validation file, sharing the same source file of functions so that no second copy can drift out of sync, checks three distinct things before the simulation is permitted to use them.

First, agreement with `psych::alpha()`. Across 360 cases drawn from this design's own data-generating process and grid, including the 69 whose item pool contained a negatively loading item, the largest absolute discrepancy in either the full-pool α or any leave-one-out α was 1.2×10^{-15}, and the two implementations nominated the same candidate item J^* in 360 of 360 cases. Cases containing a negatively loading item are included deliberately, since neither implementation applies automatic reverse keying and such pools are the case most likely to expose a difference.

Second, the closed-form population targets. α_{pop} and ρ computed from the loadings were compared against the same quantities computed independently from the model-implied population covariance matrix $\Sigma = \lambda\lambda^\top + \text{diag}(1 - \lambda_i^2)$. The largest discrepancy was 1.3×10^{-15}. The inequality $\alpha_{\text{pop}}(S) \leq \rho(S)$ that the estimands rely on held in every case.

Third, and least obviously, that the generated data actually has the population structure the targets assume. This is not a hypothetical concern. Given `lavaan` syntax specifying loadings only, `simulateData()` leaves the residual variances free and defaults them to 1, producing items of variance $\lambda_i^2 + 1$ rather than unit variance. The population targets would then describe a different population from the one being sampled, and every bias estimand in the design would be displaced by a constant that looks exactly like finite-sample bias. The residual variances are therefore stated explicitly as $\psi_i = 1 - \lambda_i^2$, and the validation confirms this worked: generating 100,000 observations from one pool at each item-pool size, the mean realised item variance departed from 1 by at most 0.0024, and the realised sample α departed from the closed-form population α by at most 0.0009, which is within Monte Carlo error at that sample size. The validation caches are keyed on a hash of the function file, so editing those functions invalidates the validation rather than allowing a stale pass to stand.

2.6. Expectations prior to running the simulation

This simulation was not preregistered. We nonetheless recorded, in the design document, what we expected each performance measure to show before the full run was executed, and we report those expectations here so that the results can be read against them rather than only in hindsight. They should be read as the authors' priors, not as a preregistered test.

We expected the full-pool calibration bias to be small and negative and to shrink toward zero as N grew, since sample α carries a known small-sample downward bias. The structural gap to be negative in every condition, since the fitted loading distribution guarantees unequal loadings. We expected to reproduce the Kopalle & Lehmann (1997) optimism result, with the reported α biased upward relative to the population α of the retained items, and we expected that bias to be larger at larger item-pool sizes, on the reasoning that more candidate items means a larger expected maximum over noise. We expected the reliability-scale bias to be negative at large N, where selection has little noise to exploit and only the structural gap remains, and possibly positive at small N, where the winner's curse might dominate - where the sign crossed, we treated as an open empirical question. We expected Pr(drop) to rise with item-pool size. We made no directional prediction for the mean population-α change or the mean reliability change, because the empirical loading distribution has a negative tail, so some pools contain a genuinely harmful item whose removal helps while others pay a pure length penalty. We expected the prediction intervals to be wide relative to the means wherever selection had room to operate. Which of these held is taken up in the Discussion.

2.7. Estimands

Because the data-generating process fixes the population loadings directly, two population quantities are available in closed form for any item set S of size m, writing $S_1 = \sum_{i \in S} \lambda_i$ and $S_2 = \sum_{i \in S} \lambda_i^2$. The population Cronbach's α is

$$\alpha_{\text{pop}}(S) = \frac{m}{m-1} \cdot \frac{S_1^2 - S_2}{m + S_1^2 - S_2}$$

and the true reliability of the unit-weighted sum score, McDonald's ω, is the proportion of sum-score variance due to the common factor:

$$\rho(S) = \frac{S_1^2}{S_1^2 + \sum_{i \in S} \psi_i} = \frac{S_1^2}{S_1^2 + m - S_2}$$

By the Cauchy-Schwarz inequality, $S_1^2 \leq m S_2$, so $\alpha_{\text{pop}}(S) \leq \rho(S)$ always, with equality only under essential tau-equivalence (all loadings in S equal). Because the fitted loading distribution produces unequal loadings, this inequality is strict in essentially every generated pool: α structurally understates true reliability even at infinite sample size, before any item selection at all. This is why a congeneric data-generating process is the appropriate one for this question, since a tau-equivalent one would collapse the two targets into one by construction.

Both population targets are compared, within the same replicate, against the corresponding sample Cronbach's α, giving an exact per-replicate decomposition of the total bias against reliability:

$$\underbrace{\hat{\alpha}(S_s) - \rho(S_s)}_{\text{total bias vs. reliability}} = \underbrace{[\hat{\alpha}(S_s) - \alpha_{\text{pop}}(S_s)]}_{\text{statistical bias (sampling + selection)}} + \underbrace{[\alpha_{\text{pop}}(S_s) - \rho(S_s)]}_{\text{structural gap} \leq 0}$$

The two components can pull in opposite directions: selection pushes the reported α up relative to its own population value, while the structural gap means that population value sits below true reliability. Whether the reported α ends up over- or under-stating true reliability, and where in the design the sign flips, is an empirical question this simulation answers.

Because J^* is a random variable across replicates, the population targets are realised (replication-specific, strategy-specific, data-dependent) estimands rather than fixed population parameters. This is deliberate: the applied question is not "is the best possible subset's reliability well estimated" but "how reliable, in truth, are the specific items this procedure just picked, and does the number I report for them tell me that?"

A second family of estimands evaluates the decision rather than the reported number:

$\widehat{\Delta}_{\text{obs}} = \hat{\alpha}(S_s) - \hat{\alpha}(F)$, the observed sample gain: the quantity the researcher sees and that, for the conditional strategy, triggers the decision. Under the forced strategy this can be negative, since the best available drop may still hurt the sample α, under the conditional strategy it is ≥ 0 by construction and exactly 0 when nothing is dropped.

$\Delta_{\text{pop}} = \alpha_{\text{pop}}(S_s) - \alpha_{\text{pop}}(F)$, the population α change for the selected scale. Removal alters the item-variance sum, the sum-score variance, and the length multiplier $\frac{m}{m-1}$ simultaneously, so the sign is a trade-off and $\Delta_{\text{pop}} \neq 0$ in general.

$\Delta_\rho = \rho(S_s) - \rho(F)$, the true reliability change, which is the quantity that actually matters for genuine measure refinement. The primary failure mode this design can detect is $\Delta_\rho < 0$ while $\widehat{\Delta}_{\text{obs}} > 0$, that is, dropping a low-consistency but valid item raises the α coefficient while degrading measurement.

The findings live in the gaps between these deltas. $\widehat{\Delta}_{\text{obs}} - \Delta_{\text{pop}}$ is the selection optimism (winner's curse), expected to be positive because J^* maximises the sample α and is therefore biased high for its own population value. It is computed pairwise within replicate, so its Monte Carlo standard error is much smaller than differencing two separate summaries would give. Because a mean Δ_ρ near zero can conceal frequent, offsetting harm, we additionally report the harm rate, $\Pr(\Delta_\rho < 0)$: the proportion of replications in which the drop actually reduced true reliability. Table 1 summarises every estimand and its role.

Table 1: Summary of the estimands. All quantities in rows 2–3 are computed per strategy; for the conditional strategy they are additionally reported conditional on a drop having occurred.

#	Quantity	Definition	Role
1a	Calibration bias (full pool)	$\mathbb{E}[\hat{\alpha}(F) - \alpha_{\text{pop}}(F)]$	pipeline check; baseline for 2a
1b	Structural gap (full pool)	$\mathbb{E}[\alpha_{\text{pop}}(F) - \rho(F)]$	≤ 0; exists with no selection
1c	Total full-pool bias	$\mathbb{E}[\hat{\alpha}(F) - \rho(F)]$	1a + 1b; no-dropping benchmark
2a	Reported-alpha bias, alpha scale	$\mathbb{E}[\hat{\alpha}(S_s) - \alpha_{\text{pop}}(S_s)]$	statistical bias incl. winner's curse

#	Quantity	Definition	Role
2b	Reported-alpha bias, reliability scale	$\mathbb{E}[\hat{\alpha}(S_s) - \rho(S_s)]$	total bias in the reported estimate
3a	Observed gain	$\mathbb{E}[\widehat{\Delta}_{\text{obs}}]$	the trigger; researcher's-eye view
3b	Population alpha change	$\mathbb{E}[\Delta_{\text{pop}}]$	did the drop improve alpha in truth
3c	True reliability change	$\mathbb{E}[\Delta_\rho]$	did the drop improve measurement
3d	Selection optimism	$\mathbb{E}[\widehat{\Delta}_{\text{obs}} - \Delta_{\text{pop}}]$	winner's curse, paired per replicate
3e	P(drop)	$\Pr(\hat{\alpha}(-J^*) > \hat{\alpha}(F))$	conditional strategy only (forced $\equiv 1$)
3f	Harm rate	$\Pr(\Delta_\rho < 0)$	how often dropping reduced reliability
3g	Harm magnitude	$\mathbb{E}[\Delta_\rho \mid \Delta_\rho < 0]$	how large the harm is when it occurs
4	Prediction interval	$[\hat{q}_{2.5}(d),\ \hat{q}_{97.5}(d)]$ for d in 2a–3d	spread, not precision: one study's range

2.8. Performance measures and Monte Carlo uncertainty

Four types of performance measure were computed per condition over its replications.

1. Calibration and structural gap (full pool, no dropping): the mean signed difference between sample α and population α, which checks the pipeline is calibrated before any selection, and the mean difference between population α and ω, which is the baseline structural gap that exists regardless of dropping.
2. Bias in the reported α of the retained items, per strategy, against both population α and true reliability, reported both marginally and, for the conditional strategy, conditional on a drop having occurred.
3. Impact of item dropping: the expected change from full pool to retained items on all three scales, the selection-optimism gap, the probability that the conditional strategy drops at all, and the harm rate.
4. 95% prediction intervals (empirical 2.5th and 97.5th percentiles across replications) for every quantity in 2 and 3, reported alongside their means, describing the spread an individual study can encounter, which the mean alone does not.

All continuous estimands are means of a per-replicate difference. Writing d_k for the relevant difference in replication k of a condition, for example $\hat{\alpha}(S_s) - \rho(S_s)$ for the headline reliability-bias estimand,

$$\widehat{\mathbb{E}}[d] = \frac{1}{K}\sum_{k=1}^{K} d_k, \qquad \widehat{\text{MCSE}}(\widehat{\mathbb{E}}[d]) = \frac{S_d}{\sqrt{K}}$$

where S_d is the standard deviation of d_k across the K replications. In the compendium these are computed with `simhelpers::calc_absolute()` (Joshi & Pustejovsky, 2022), which implements the Siepe et al. (2024) bias and Monte Carlo standard error formulas. That function requires a single fixed true parameter per condition, whereas here every replication draws its own loading vector and therefore has its own target. We therefore pass the per-replicate difference d_k itself as the estimate with a true parameter of 0, which is algebraically identical to bias against a varying target. Proportions take the binomial Monte Carlo standard error, $\widehat{\text{MCSE}}(\widehat{p}) = \sqrt{\widehat{p}(1-\widehat{p})/K}$.

For the conditional strategy, all measure-2 and measure-3 quantities are additionally summarised over only the replications in which a drop occurred: the number of such replications, K_{drop}, is itself random and is reported alongside every conditional-on-drop estimate. For measure 4 we report the empirical 2.5th and 97.5th percentiles of d_k,

$$\widehat{\text{PI}}_{95}(d) = [\hat{q}_{2.5}(d_k),\ \hat{q}_{97.5}(d_k)]$$

The prediction-interval bounds are themselves Monte Carlo estimates and carry their own uncertainty, which cannot be assumed negligible: unlike a mean, a sample quantile's standard error depends on the density of the distribution at that quantile, $\text{SE}(\hat{q}_p) \approx \sqrt{p(1-p)/K} \,/\, f(q_p)$, so it can be large wherever the tail is thin, and quantile estimates in heavy-tailed or sparse regions can be unstable. We therefore estimated a Monte Carlo standard error for every reported bound using the exact distribution-free order-statistic method (the 95% confidence interval for a population quantile lies between the order statistics whose ranks are the Binomial(K, p) percentiles), cross-checked against a nonparametric bootstrap, with which it agrees to within roughly 30% in every cell. The bounds turn out to be stable: across all 300 cell-by-quantity combinations, the larger of the two bound standard errors is at most 2.2% of its interval's width (median 1.2%), so every interval in Table 7 and Figure 5 is accurate to well within the precision at which it is printed. As expected, quantiles are estimated less precisely than means, by a factor of up to 8 in the same cell, and the least stable bounds are the upper tails of the observed-gain and optimism distributions at the smallest sample size, which are the heaviest tails in the design. One degenerate case is worth noting: for the conditional strategy the lower bound of the observed gain sits on the point mass at zero contributed by the no-drop replications, so that bound has a Monte Carlo standard error of exactly zero, an artefact of the mixture rather than evidence of precision, and is another reason the conditional strategy's interval is more informative conditional on a drop having occurred.

Monte Carlo standard errors and prediction intervals are different quantities and should not be confused. An MCSE is $S_d/\sqrt{K}$ and describes how precisely the mean is estimated, decreasing as the number of iterations (K) grows. A prediction interval is driven by S_d itself and describes how much individual studies differ; as K grows its *estimate* stabilises but its *width* does not shrink, because the width is a property of the distribution rather than of how precisely we have sampled it. At $K = 10{,}000$ the MCSE is roughly one hundredth of the standard deviation, which is precisely why Aim 4 requires its own summaries rather than wider error bars on the existing ones.

Because each replication draws both a fresh item pool and a fresh sample from it, the prediction intervals marginalise over two sources of variation at once. They therefore describe the range across studies of psychological trait scales in general, which is the right quantity for a field-level claim, and not the range available to one researcher holding one fixed scale, which would require a nested design. Simulations were parallelised with reproducible L'Ecuyer-CMRG streams, so results are identical regardless of the number of workers used.

Table 2: Full item pool, no dropping. Estimate (+/-1 Monte Carlo standard error) of the calibration bias of sample alpha against population alpha (1a), the structural gap between population alpha and true reliability (1b), and their sum (1c), by item-pool size (k) and sample size (N).

k	N	1a. Calibration bias	1b. Structural gap	1c. Total bias
5	20	-0.0423 (0.0018)	-0.0242 (0.0002)	-0.0665 (0.0018)
	30	-0.0293 (0.0014)	-0.0244 (0.0002)	-0.0537 (0.0014)
	50	-0.0167 (0.0010)	-0.0240 (0.0002)	-0.0407 (0.0010)
	200	-0.0041 (0.0005)	-0.0240 (0.0002)	-0.0282 (0.0005)
	1000	-0.0008 (0.0002)	-0.0240 (0.0002)	-0.0248 (0.0003)
10	20	-0.0262 (0.0010)	-0.0144 (0.0001)	-0.0406 (0.0010)
	30	-0.0178 (0.0008)	-0.0144 (0.0001)	-0.0322 (0.0008)
	50	-0.0090 (0.0006)	-0.0142 (0.0001)	-0.0232 (0.0006)
	200	-0.0028 (0.0003)	-0.0143 (0.0001)	-0.0171 (0.0003)
	1000	-0.0004 (0.0001)	-0.0145 (0.0001)	-0.0149 (0.0001)
15	20	-0.0189 (0.0007)	-0.0102 (0.0001)	-0.0292 (0.0007)
	30	-0.0128 (0.0005)	-0.0102 (0.0001)	-0.0230 (0.0005)
	50	-0.0070 (0.0004)	-0.0103 (0.0001)	-0.0173 (0.0004)
	200	-0.0015 (0.0002)	-0.0102 (0.0001)	-0.0117 (0.0002)
	1000	-0.0003 (0.0001)	-0.0103 (0.0001)	-0.0106 (0.0001)
20	20	-0.0145 (0.0005)	-0.0079 (0.0000)	-0.0225 (0.0005)
	30	-0.0090 (0.0004)	-0.0079 (0.0000)	-0.0169 (0.0004)
	50	-0.0050 (0.0003)	-0.0079 (0.0000)	-0.0129 (0.0003)
	200	-0.0013 (0.0001)	-0.0079 (0.0000)	-0.0091 (0.0001)
	1000	-0.0002 (0.0001)	-0.0079 (0.0000)	-0.0081 (0.0001)
40	20	-0.0074 (0.0003)	-0.0041 (0.0000)	-0.0115 (0.0003)
	30	-0.0045 (0.0002)	-0.0041 (0.0000)	-0.0086 (0.0002)
	50	-0.0025 (0.0001)	-0.0041 (0.0000)	-0.0067 (0.0001)
	200	-0.0007 (0.0001)	-0.0041 (0.0000)	-0.0048 (0.0001)
	1000	-0.0001 (0.0000)	-0.0041 (0.0000)	-0.0042 (0.0000)

3. Results

Results are organised around the four performance measures. Every table reports all 25 conditions of the design rather than a selected or pooled subset, because the estimands vary substantially and non-monotonically across the grid and an average over the grid is an average over the design chosen here rather than over any population of real studies.

3.1. Calibration and the structural gap

Estimating the full-pool Cronbach's α shows a small downward calibration bias at small samples (Table 2, column 1a; Figure 1, panel 1a). With small samples, the item covariances that α is built from are estimated with high sampling noise. The bias decreases toward zero as N grows at every item-pool size, from -0.0423 at $k = 5$, $N = 20$ to -0.0008 at $N = 1000$, and is smaller for larger item pools even at fixed N (-0.0074 at $k = 40$, $N = 20$).

The structural gap between population α and true reliability behaves quite differently. It is a property of the loadings alone and does not depend on the sample at all, so its estimate is essentially invariant to N: within a fixed item-pool size it varies by no more than 0.0004 across the full $N = 20$ to 1000 range, a residual difference attributable to Monte Carlo error in the loadings drawn rather than to sample size. The gap does shrink as pool size increases, from -0.0241 at $k = 5$ to -0.0041 at $k = 40$, because both population α and true reliability approach a ceiling near 1 as scales lengthen, compressing the distance between them even though the underlying inequality among item loadings is unchanged.

These two components sum to the total full-pool bias (column 1c), which is negative everywhere: with no item dropping at all, the reported α understates the true reliability of the scale in every condition examined, by between 0.0042 and 0.0665.

Table 3: Bias (+/-1 Monte Carlo standard error) of the reported (sample) alpha of the retained items against the true population alpha of those same items (2a) and against their true reliability (2b), by strategy, item-pool size (k), and sample size (N). The forced strategy drops in every replication by definition. Conditional-strategy columns are marginal, so replications in which nothing was dropped contribute the full-pool values.

		Forced		Conditional (marginal)	
k	N	2a vs. pop. alpha	2b vs. reliability	2a vs. pop. alpha	2b vs. reliability
5	20	0.0435 (0.0013)	0.0239 (0.0013)	0.0415 (0.0013)	0.0220 (0.0012)
	30	0.0315 (0.0011)	0.0131 (0.0010)	0.0293 (0.0010)	0.0110 (0.0010)
	50	0.0228 (0.0008)	0.0059 (0.0008)	0.0211 (0.0008)	0.0042 (0.0008)
	200	0.0064 (0.0004)	-0.0084 (0.0004)	0.0057 (0.0004)	-0.0091 (0.0004)
	1000	0.0015 (0.0002)	-0.0123 (0.0002)	0.0012 (0.0002)	-0.0125 (0.0002)
10	20	0.0057 (0.0008)	-0.0062 (0.0008)	0.0056 (0.0008)	-0.0063 (0.0008)
	30	0.0045 (0.0006)	-0.0069 (0.0006)	0.0044 (0.0006)	-0.0069 (0.0006)
	50	0.0048 (0.0005)	-0.0058 (0.0005)	0.0048 (0.0005)	-0.0059 (0.0005)
	200	0.0009 (0.0002)	-0.0090 (0.0002)	0.0008 (0.0002)	-0.0090 (0.0002)
	1000	0.0003 (0.0001)	-0.0093 (0.0001)	0.0003 (0.0001)	-0.0093 (0.0001)
15	20	-0.0017 (0.0006)	-0.0103 (0.0006)	-0.0017 (0.0006)	-0.0103 (0.0006)
	30	-0.0008 (0.0004)	-0.0092 (0.0004)	-0.0008 (0.0004)	-0.0092 (0.0004)
	50	0.0003 (0.0003)	-0.0078 (0.0003)	0.0003 (0.0003)	-0.0078 (0.0003)
	200	0.0005 (0.0002)	-0.0071 (0.0002)	0.0005 (0.0002)	-0.0071 (0.0002)
	1000	0.0001 (0.0001)	-0.0073 (0.0001)	0.0001 (0.0001)	-0.0073 (0.0001)
20	20	-0.0035 (0.0005)	-0.0103 (0.0005)	-0.0035 (0.0005)	-0.0103 (0.0005)
	30	-0.0015 (0.0003)	-0.0081 (0.0003)	-0.0015 (0.0003)	-0.0081 (0.0003)
	50	-0.0004 (0.0003)	-0.0068 (0.0003)	-0.0004 (0.0003)	-0.0068 (0.0003)
	200	-0.0000 (0.0001)	-0.0061 (0.0001)	-0.0000 (0.0001)	-0.0061 (0.0001)
	1000	0.0000 (0.0001)	-0.0059 (0.0001)	0.0000 (0.0001)	-0.0059 (0.0001)
40	20	-0.0039 (0.0002)	-0.0076 (0.0002)	-0.0039 (0.0002)	-0.0076 (0.0002)
	30	-0.0021 (0.0002)	-0.0058 (0.0002)	-0.0021 (0.0002)	-0.0058 (0.0002)
	50	-0.0010 (0.0001)	-0.0046 (0.0001)	-0.0010 (0.0001)	-0.0046 (0.0001)
	200	-0.0003 (0.0001)	-0.0038 (0.0001)	-0.0003 (0.0001)	-0.0038 (0.0001)
	1000	0.0000 (0.0000)	-0.0034 (0.0000)	0.0000 (0.0000)	-0.0034 (0.0000)

3.2. Bias in the reported alpha after item dropping

Against the population α of the retained items (Table 3, column 2a), the selection-induced optimism predicted by Kopalle & Lehmann (1997) is clearly present at the shortest scales, where the reported α overstates the population α of the very items it describes by 0.0435 at $k = 5$, $N = 20$. It shrinks rapidly with both factors, and at $k \geq 15$ with small samples it turns slightly negative, because the small-sample downward calibration bias of α documented above is then larger than the optimism the argmax can extract from a pool that is already long enough for α to be estimated fairly stably. Selection optimism is therefore not uniformly larger in longer scales, contrary to the intuition that

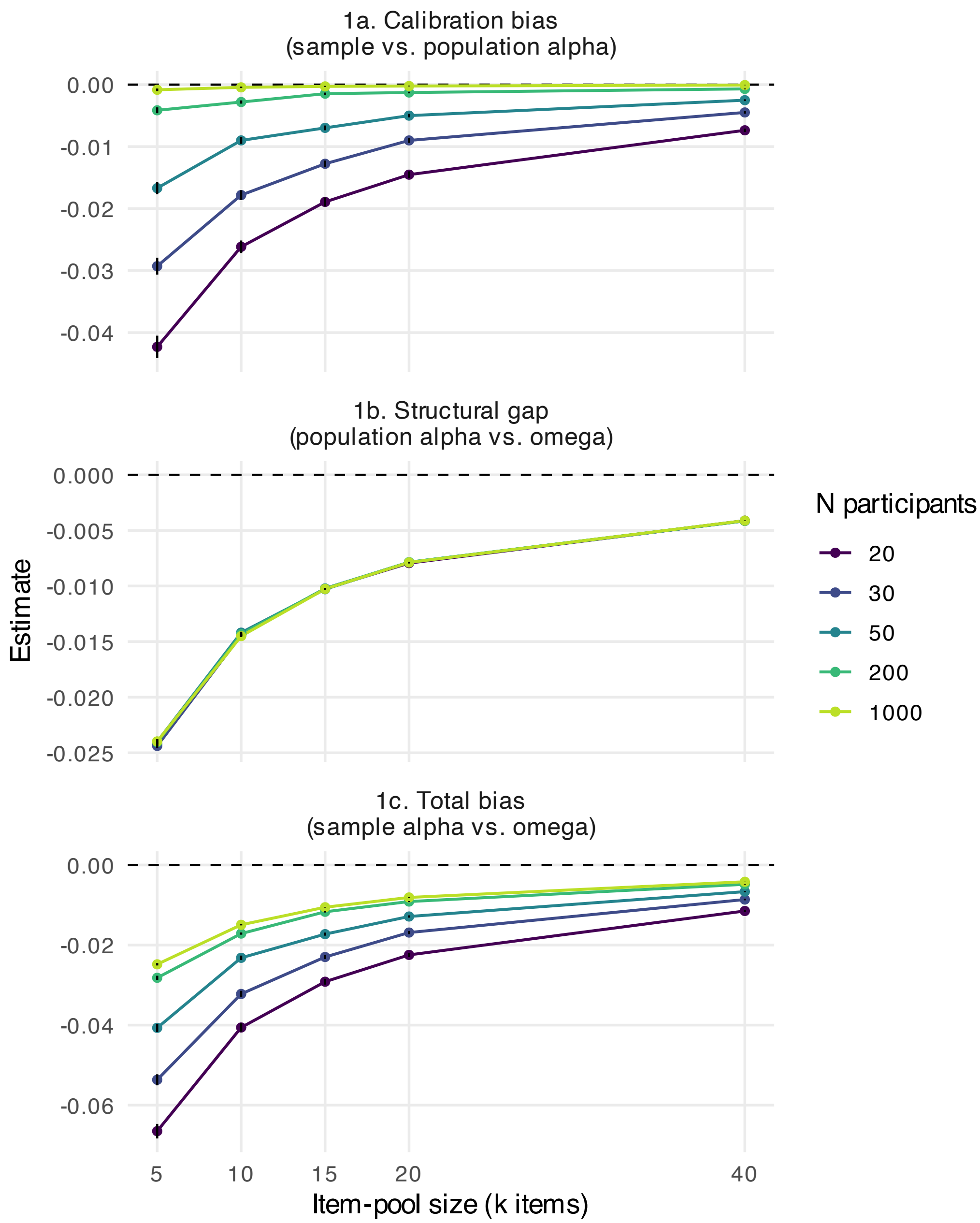

Figure 1: Full-pool calibration bias (1a), structural gap between population alpha and omega (1b), and their sum, the total bias against true reliability with no item dropping (1c), by item-pool size (x axis) and sample size (colour). Error bars are +/-1 Monte Carlo standard error and are narrower than the plotting symbol in most cells. Note the differing y-axis scales. The structural gap (1b) depends only on the loadings, so its five sample-size series coincide.

more candidate items means more opportunity to capitalise on noise. With k items there are more candidates, but each contributes proportionally less to α, and the second effect dominates.

Against true reliability (column 2b), which is the applied target because α is reported as a reliability estimate, the two components combine. The reported α overstates true reliability only in the shortest scales at small samples, in 3 of the 25 conditions, all at $k = 5$: by 0.0239 at $N = 20$, falling to 0.0059 at $N = 50$. In every other condition it understates true reliability, by between 0.003 and 0.012. The sign of the bias in a reported α after item dropping is therefore not fixed: it depends on where a given study sits relative to selection optimism on one side and small-sample calibration bias plus the structural gap on the other.

The same two estimands can be shown on the scale an applied reader thinks in, which is also the form in which Kopalle & Lehmann (1997) summarised their own results and is therefore the directly comparable presentation: the true value of the retained items on the horizontal axis, the reported sample α of those same items on the vertical axis, with the identity line as the reference (Figure 3). Vertical distance above the identity line is exactly the bias of estimand 2a (top row) or 2b (bottom row), and a no-dropping series is included so that the effect of selection is visible against the baseline of not selecting at all. Because every replication draws its own loadings, the true value varies continuously within a condition rather than being fixed, so replications are binned by it at a width of 0.05 and the mean reported α is plotted per bin. Bins containing fewer than 20 replications are not plotted, which removes 0.05% of replications in the sparse tails of the true-value distribution.

Two features are worth reading off it. First, the no-dropping series does not lie on the identity line: against the population α of the retained items (top row) it sits below it, by 0.010 on average where the true value exceeds 0.45 and by 0.025 where the true value is lower. This is the small-sample calibration bias seen from a different angle, and it is largest at low true values because those are the short, weakly loaded pools whose covariances are estimated least precisely. Any assessment of what selection does has to be read against that sloping baseline rather than against the identity line.

Second, both dropping series sit above the no-dropping series everywhere, and the vertical distance between them is the selection effect in its purest form, since the three series differ only in whether and how an item was chosen. At high true values that shift roughly cancels the calibration bias, leaving the reported α close to correct (0.001 for the forced rule). At low true values it substantially overshoots, leaving the reported α 0.070 above the truth. This is the Kopalle and Lehmann result in its original form, and it makes clear that their "alpha inflation" is not a uniform upward shift but something that bites hardest exactly where reliability is already poor.

The bottom row is the same replications compared against true reliability instead. Every series moves down by the structural gap, which is why the no-dropping series now sits 0.072 below the identity line at low true values. Dropping partly offsets that displacement rather than compounding it, which is the graphical form of the sign reversal in Table 3: at high true values the forced rule lands 0.007 below the truth, and only at low true values does it end up above it (0.041).

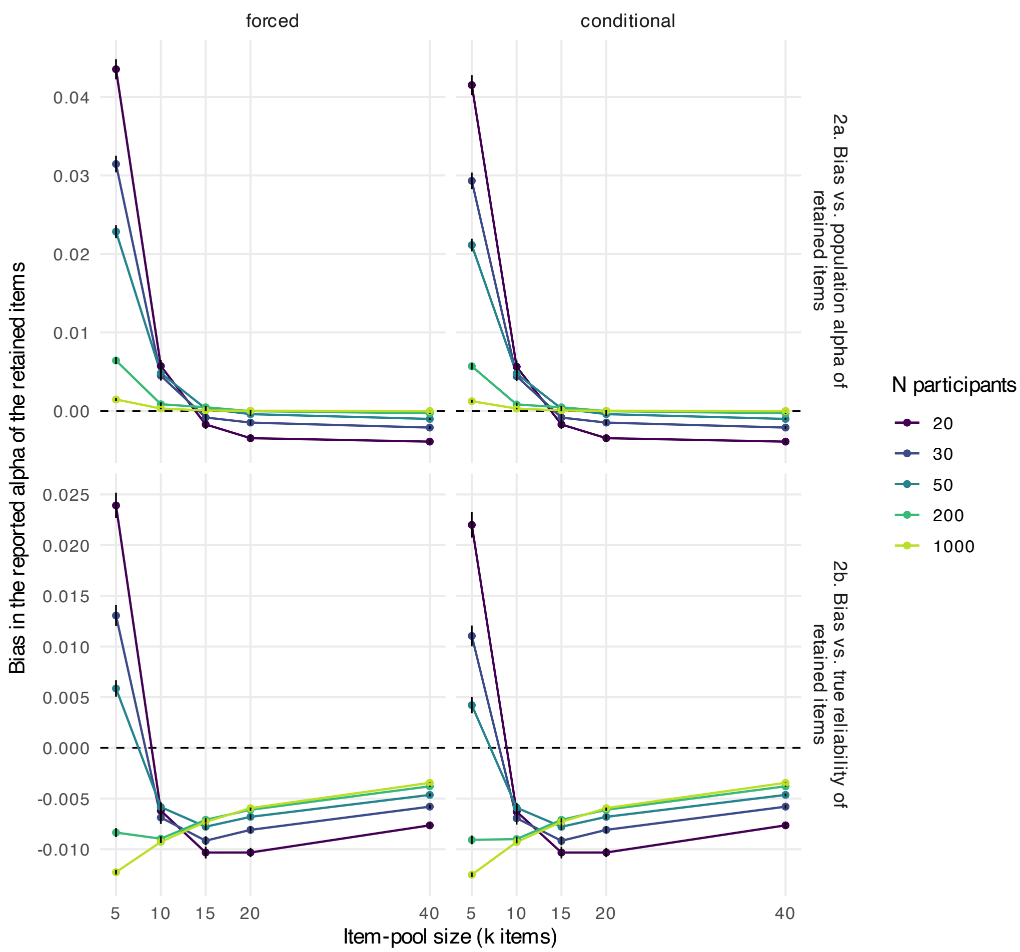

Figure 2: Bias in the reported alpha of the retained items against the population alpha of those items (top row, 2a) and against their true reliability (bottom row, 2b), for the forced (left) and conditional (right) strategies, by item-pool size and sample size. Error bars are +/-1 Monte Carlo standard error. The two strategies are near-identical in every panel, consistent with their shared nomination rule. Note the differing y-axis scales between rows.

These curves must not be read backwards. They give the expected reported α given the truth. The question an applied researcher faces is the reverse one, since they know their reported α and want the truth, but the expected true value given a reported α is a different quantity that these curves do not provide, as it depends also on how often each true value arises in the first place. Reading horizontally from a reported α across to a curve would overstate what can be inferred from it.

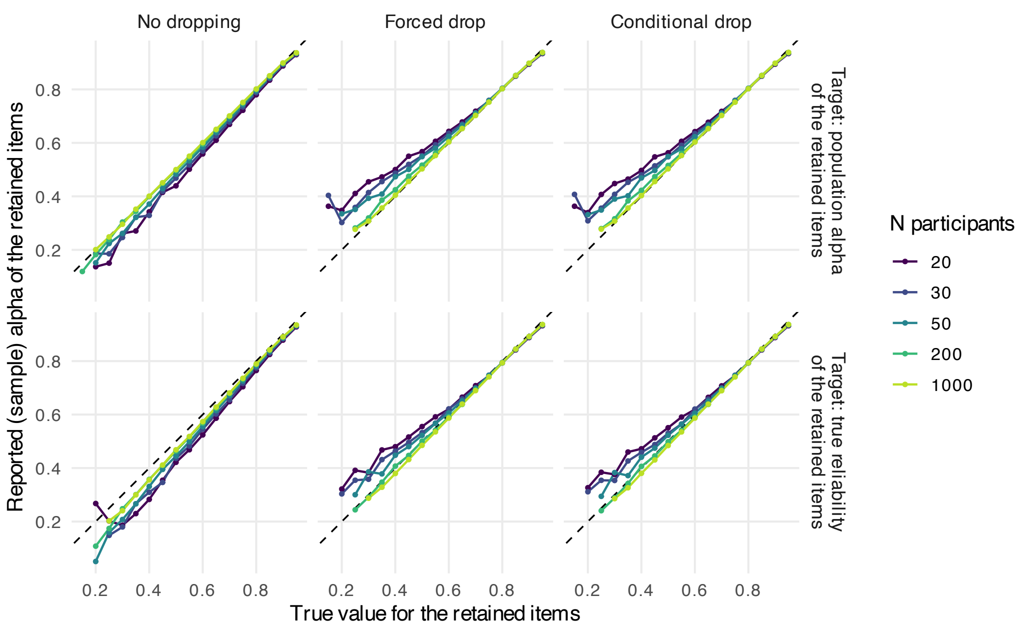

Figure 3: Calibration view: the reported (sample) alpha of the retained items against their true value, in the presentation used by Kopalle and Lehmann (1997). Columns are the two dropping strategies plus a no-dropping reference, rows are the two population targets. The dashed line is the identity: points on it mean the reported alpha is exactly right for the items retained, and points above it mean it overstates them. Replications are binned by their true value at a width of 0.05 and averaged within bin. Bins with fewer than 20 replications are omitted. Binning is presentational only and no estimand is defined in terms of it.

Table 4: Conditional strategy, restricted to the replications in which a drop actually occurred. Estimate (+/-1 Monte Carlo standard error) of the bias in the reported alpha against population alpha (2a) and against true reliability (2b), the mean true reliability change (3c), and the harm rate among these drops (3f; percent). K(drop) is the realised number of replications in which a drop occurred, out of 10,000. The given-drop harm rate has a different denominator from the marginal harm rate in the impact table below, which counts no-drop replications as non-harm.

k	N	K(drop)	2a vs. pop. alpha	2b vs. reliability	3c. Reliability change	3f. Harm given drop
5	20	9,102	0.0391 (0.0014)	0.0190 (0.0013)	0.0023 (0.0005)	46.7 (0.52)
	30	8,868	0.0282 (0.0011)	0.0091 (0.0011)	0.0102 (0.0005)	39.4 (0.52)
	50	8,560	0.0209 (0.0009)	0.0030 (0.0009)	0.0167 (0.0004)	33.1 (0.51)
	200	8,013	0.0054 (0.0005)	-0.0109 (0.0005)	0.0296 (0.0004)	16.0 (0.41)
	1000	7,794	0.0013 (0.0002)	-0.0143 (0.0003)	0.0339 (0.0003)	7.2 (0.29)
10	20	9,874	0.0052 (0.0008)	-0.0067 (0.0008)	0.0067 (0.0002)	32.0 (0.47)
	30	9,838	0.0041 (0.0006)	-0.0074 (0.0006)	0.0089 (0.0002)	26.9 (0.45)
	50	9,750	0.0045 (0.0005)	-0.0063 (0.0005)	0.0110 (0.0001)	19.8 (0.40)
	200	9,641	0.0007 (0.0002)	-0.0093 (0.0002)	0.0146 (0.0001)	7.6 (0.27)
	1000	9,614	0.0003 (0.0001)	-0.0095 (0.0001)	0.0159 (0.0001)	2.5 (0.16)
15	20	9,982	-0.0017 (0.0006)	-0.0104 (0.0006)	0.0051 (0.0001)	25.0 (0.43)
	30	9,978	-0.0009 (0.0004)	-0.0092 (0.0004)	0.0062 (0.0001)	19.0 (0.39)
	50	9,970	0.0003 (0.0003)	-0.0078 (0.0003)	0.0073 (0.0001)	12.8 (0.33)
	200	9,939	0.0005 (0.0002)	-0.0071 (0.0002)	0.0088 (0.0001)	3.7 (0.19)
	1000	9,919	0.0000 (0.0001)	-0.0073 (0.0001)	0.0096 (0.0001)	0.8 (0.09)
20	20	9,999	-0.0035 (0.0005)	-0.0103 (0.0005)	0.0036 (0.0000)	20.4 (0.40)
	30	9,998	-0.0015 (0.0003)	-0.0081 (0.0003)	0.0043 (0.0000)	15.2 (0.36)
	50	9,993	-0.0004 (0.0003)	-0.0068 (0.0003)	0.0050 (0.0000)	8.9 (0.28)
	200	9,989	-0.0000 (0.0001)	-0.0061 (0.0001)	0.0059 (0.0000)	1.8 (0.13)
	1000	9,985	0.0000 (0.0001)	-0.0060 (0.0001)	0.0063 (0.0000)	0.3 (0.06)
40	20	10,000	-0.0039 (0.0002)	-0.0076 (0.0002)	0.0014 (0.0000)	12.0 (0.33)
	30	10,000	-0.0021 (0.0002)	-0.0058 (0.0002)	0.0016 (0.0000)	7.0 (0.26)
	50	10,000	-0.0010 (0.0001)	-0.0046 (0.0001)	0.0018 (0.0000)	3.2 (0.18)
	200	10,000	-0.0003 (0.0001)	-0.0038 (0.0001)	0.0021 (0.0000)	0.1 (0.04)
	1000	10,000	0.0000 (0.0000)	-0.0034 (0.0000)	0.0022 (0.0000)	0.0 (0.00)

Restricting the conditional strategy to the replications in which it actually acted (Table 4) leaves this pattern almost unchanged, because both strategies remove the same nominated item whenever the conditional strategy drops at all. The realised K_{drop} shrinks furthest at the smallest pool size, where the conditional rule declines to act increasingly often as N grows, from 9.0% of replications at $N = 20$ to 22.1% at $N = 1000$. This is consistent with greater sampling precision at larger N leaving less noise for the argmax to exploit.

3.3. Impact of item dropping

Whether an item is dropped at all, and whether doing so helps or harms the scale, are both determined by the number of items and the sample size (Table 5, Figure 4). The propensity of the conditional strategy to drop an item is highest exactly where its judgment is least trustworthy. At the smallest pool size, Pr(drop) falls from 91.0% at $N = 20$ to 77.9% at $N = 1000$. At the largest pool size it remains at or near 100% regardless of sample size: with more items available there is almost always some single item whose removal raises the sample α, whether or not that increase reflects a real improvement to the scale.

Table 5: Proportion of replications in which the conditional strategy dropped an item (3e; the forced strategy drops in all replications by definition), the proportion in which the drop reduced true reliability (3f, the harm rate), and the mean size of that reduction when it occurred (3g, harm magnitude; forced strategy), by item-pool size (k) and sample size (N). Proportions are percentages with +/-1 Monte Carlo standard error in percentage points; conditional-strategy harm is marginal (no-drop replications count as non-harm; see the conditional-on-drop table for the given-drop rate). The final column is the proportion of forced-strategy replications in which the apparent in-sample gain was not positive, i.e. in which the drop did not even appear to help. An em-dash marks the one cell with no harmful replications, where the conditional mean does not exist.

k	N	P(drop)	Harm, forced	Harm, cond.	3g. Mean harm	No gain, forced
5	20	91.0 (0.29)	49.6 (0.50)	42.5 (0.49)	-0.0398 (0.0005)	9.0
	30	88.7 (0.32)	43.9 (0.50)	34.9 (0.48)	-0.0348 (0.0005)	11.3
	50	85.6 (0.35)	39.3 (0.49)	28.3 (0.45)	-0.0290 (0.0004)	14.4
	200	80.1 (0.40)	28.9 (0.45)	12.8 (0.33)	-0.0182 (0.0003)	19.9
	1000	77.9 (0.41)	25.3 (0.43)	5.6 (0.23)	-0.0137 (0.0002)	22.1
10	20	98.7 (0.11)	32.7 (0.47)	31.6 (0.47)	-0.0107 (0.0002)	1.3
	30	98.4 (0.13)	27.6 (0.45)	26.5 (0.44)	-0.0087 (0.0002)	1.6
	50	97.5 (0.16)	21.1 (0.41)	19.4 (0.40)	-0.0072 (0.0001)	2.5
	200	96.4 (0.19)	10.0 (0.30)	7.4 (0.26)	-0.0040 (0.0001)	3.6
	1000	96.1 (0.19)	5.6 (0.23)	2.4 (0.15)	-0.0026 (0.0001)	3.9
15	20	99.8 (0.04)	25.1 (0.43)	24.9 (0.43)	-0.0046 (0.0001)	0.2
	30	99.8 (0.05)	19.1 (0.39)	19.0 (0.39)	-0.0037 (0.0001)	0.2
	50	99.7 (0.05)	13.0 (0.34)	12.8 (0.33)	-0.0031 (0.0001)	0.3
	200	99.4 (0.08)	4.1 (0.20)	3.6 (0.19)	-0.0015 (0.0001)	0.6
	1000	99.2 (0.09)	1.4 (0.12)	0.8 (0.09)	-0.0011 (0.0001)	0.8
20	20	100.0 (0.01)	20.4 (0.40)	20.4 (0.40)	-0.0025 (0.0000)	0.0
	30	100.0 (0.01)	15.2 (0.36)	15.2 (0.36)	-0.0021 (0.0000)	0.0
	50	99.9 (0.03)	8.9 (0.28)	8.9 (0.28)	-0.0016 (0.0000)	0.1
	200	99.9 (0.03)	1.9 (0.14)	1.8 (0.13)	-0.0008 (0.0001)	0.1
	1000	99.9 (0.04)	0.4 (0.07)	0.3 (0.06)	-0.0005 (0.0001)	0.1
40	20	100.0 (0.00)	12.0 (0.33)	12.0 (0.33)	-0.0006 (0.0000)	0.0
	30	100.0 (0.00)	7.0 (0.26)	7.0 (0.26)	-0.0005 (0.0000)	0.0
	50	100.0 (0.00)	3.2 (0.18)	3.2 (0.18)	-0.0004 (0.0000)	0.0
	200	100.0 (0.00)	0.1 (0.04)	0.1 (0.04)	-0.0002 (0.0000)	0.0
	1000	100.0 (0.00)	0.0 (0.00)	0.0 (0.00)	—	0.0

The consequences of removing the item are given by the harm rate, the proportion of replications in which the drop reduced true reliability. At the smallest scale and sample size the forced strategy's harm rate is 49.6%: the nominated item is very nearly as likely to be an artefact of that particular sample as it is to be a genuine weak point in the scale. Harm falls steeply as either factor increases, but it does not vanish with sample size alone. At $N = 1000$ the forced strategy still reduces true reliability in 25.3% of replications at $k = 5$, and the rate reaches 0.0% only in the largest scale and sample combination. Long scales and large samples are both needed - neither alone is sufficient.

The harm rate counts sign only, so it must be read alongside the harm *magnitude*, the mean reliability loss among the replications in which harm occurred (Table 5, column 3g). The two together show that the harm rates are not comparable across scale lengths without the magnitudes. At $k = 5$, $N = 20$, harm is both frequent and material: the mean loss given harm is -0.040, comparable in size to the mean *gain* among helpful drops. At $k = 40$ the harm rate of 12.0% at $N = 20$ counts losses averaging only -0.0006, trivial beside those at short scales. The practical risk is therefore concentrated where the rate and the magnitude are large together, in short scales at small samples.

The conditional strategy's harm rates require care with denominators, because two are available and they answer different questions. Marginally, across all replications including those where nothing was dropped, the conditional rule harms in 42.5% of replications at $k = 5$, $N = 20$, versus 49.6% under the forced rule. But conditional on the rule actually firing, which is the risk a researcher who has just been told to drop an item faces, the harm rate is 46.7% (Table 4): barely below the forced rule's. Declining to act in the replications with the weakest apparent improvement lowers the incidence of harm across all studies mostly by gambling less often, not by gambling better. The gating rule is a filter on how often the gamble is taken, and only weakly on whether it pays off.

An important qualification concerns how the harm rate should be described. It is tempting to say the drop harms reliability "while appearing to improve α every time", but this holds only for the conditional strategy, for which $\widehat{\Delta}_{\text{obs}} > 0$ is true by construction whenever it acts. The forced strategy has no such guarantee: its apparent gain was not positive in 9.0% of replications at $k = 5$, $N = 20$, rising to 22.1% at $N = 1000$ (Table 5, final column). The strongest claim the design supports without qualification is therefore the conditional one: at $k = 5$, $N = 20$, the rule that acts only when the in-sample α improves nevertheless reduced true reliability in 46.7% of the replications in which it acted.

The same pattern appears in the magnitude of what item dropping does to the scale (Table 6). The observed sample gain $\widehat{\Delta}_{\text{obs}}$ is positive on average in every condition, including those with the highest harm rates. The population α change Δ_{pop} and the true reliability change Δ_ρ are both consistently smaller, and the gap between $\widehat{\Delta}_{\text{obs}}$ and Δ_{pop}, the selection optimism, is largest exactly where the harm rate is highest: 0.086 at $k = 5$, $N = 20$, falling to 0.0001 at $k = 40$, $N = 1000$. The conditions producing the most inflated apparent gain are the same conditions producing the highest probability that the underlying decision was actively counterproductive. Most strikingly, at $k = 5$, $N = 20$ the forced rule's mean apparent gain of 0.090 in sample α corresponds to a mean change in true reliability of -0.0007, which is to say essentially none at all: almost the entire apparent improvement is selection optimism.

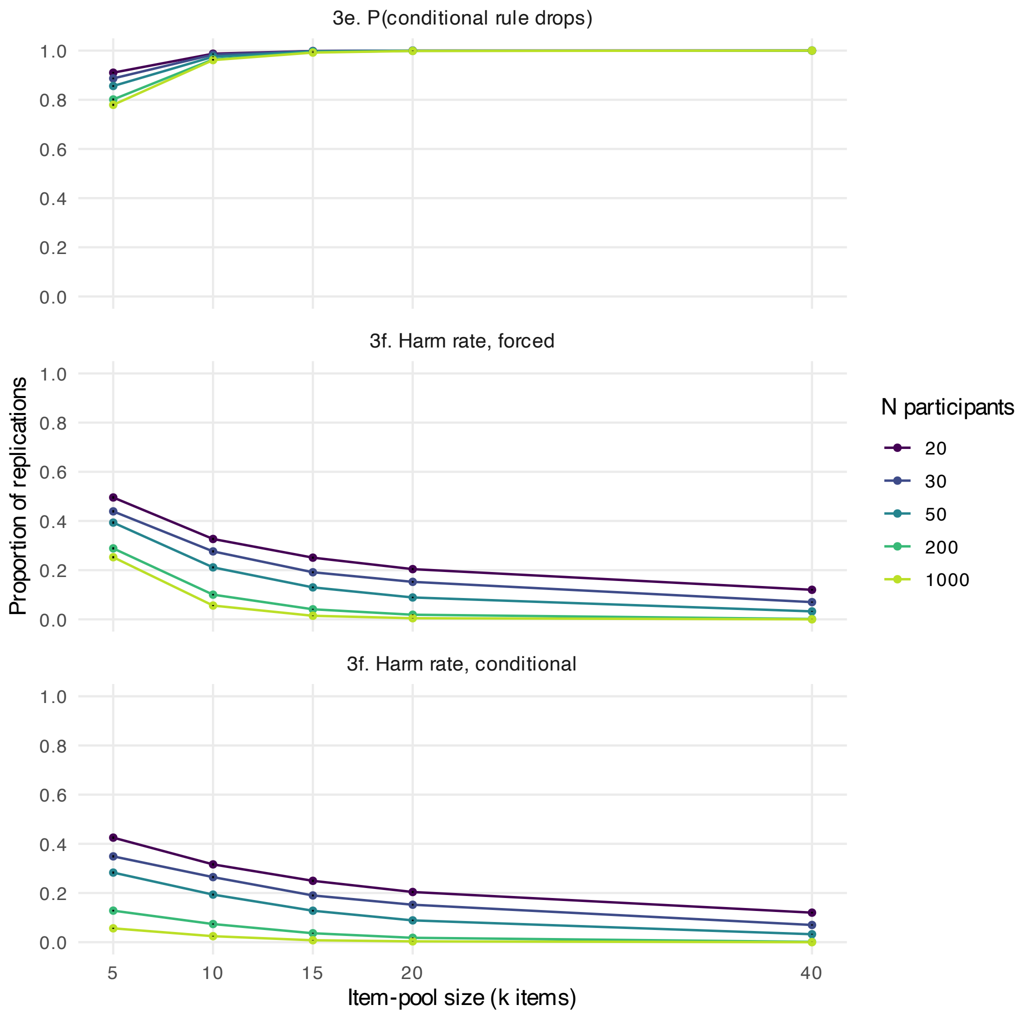

Figure 4: Probability that the conditional strategy drops an item (left) and the probability that the drop reduced true reliability, i.e. the harm rate, for each strategy (centre and right), by item-pool size and sample size. Error bars are +/-1 Monte Carlo standard error. The conditions in which the conditional rule is most likely to act are the conditions in which acting is most likely to harm the scale.

Table 6: Mean change (+/-1 Monte Carlo standard error) from full pool to retained items on three scales: observed sample alpha (3a), population alpha (3b), and true reliability (3c), together with the selection optimism (3d = 3a - 3b, computed pairwise within replicate), for the forced strategy, by item-pool size (k) and sample size (N). Conditional-strategy values are given in the compendium, they differ only through the replications in which no drop occurred, which contribute zeros.

k	N	3a. Observed	3b. Population alpha	3c. True reliability	3d. Optimism
5	20	0.0897 (0.0011)	0.0038 (0.0006)	-0.0007 (0.0005)	0.0858 (0.0010)
	30	0.0726 (0.0008)	0.0118 (0.0006)	0.0058 (0.0005)	0.0607 (0.0008)
	50	0.0576 (0.0007)	0.0180 (0.0006)	0.0110 (0.0004)	0.0395 (0.0006)
	200	0.0404 (0.0005)	0.0298 (0.0005)	0.0206 (0.0003)	0.0106 (0.0003)
	1000	0.0359 (0.0005)	0.0336 (0.0005)	0.0234 (0.0003)	0.0023 (0.0001)
10	20	0.0409 (0.0004)	0.0090 (0.0002)	0.0065 (0.0002)	0.0319 (0.0004)
	30	0.0340 (0.0003)	0.0116 (0.0002)	0.0086 (0.0002)	0.0223 (0.0003)
	50	0.0279 (0.0002)	0.0141 (0.0002)	0.0106 (0.0001)	0.0138 (0.0002)
	200	0.0221 (0.0002)	0.0184 (0.0002)	0.0139 (0.0001)	0.0037 (0.0001)
	1000	0.0209 (0.0002)	0.0201 (0.0002)	0.0152 (0.0001)	0.0007 (0.0000)
15	20	0.0239 (0.0002)	0.0067 (0.0001)	0.0051 (0.0001)	0.0172 (0.0002)
	30	0.0200 (0.0001)	0.0080 (0.0001)	0.0061 (0.0001)	0.0119 (0.0001)
	50	0.0167 (0.0001)	0.0095 (0.0001)	0.0072 (0.0001)	0.0073 (0.0001)
	200	0.0134 (0.0001)	0.0115 (0.0001)	0.0088 (0.0001)	0.0019 (0.0000)
	1000	0.0127 (0.0001)	0.0124 (0.0001)	0.0095 (0.0001)	0.0003 (0.0000)
20	20	0.0158 (0.0001)	0.0047 (0.0001)	0.0036 (0.0000)	0.0111 (0.0001)
	30	0.0131 (0.0001)	0.0056 (0.0001)	0.0043 (0.0000)	0.0075 (0.0001)
	50	0.0110 (0.0001)	0.0064 (0.0001)	0.0050 (0.0000)	0.0046 (0.0001)
	200	0.0090 (0.0001)	0.0077 (0.0000)	0.0059 (0.0000)	0.0012 (0.0000)
	1000	0.0085 (0.0000)	0.0082 (0.0000)	0.0063 (0.0000)	0.0002 (0.0000)
40	20	0.0052 (0.0000)	0.0017 (0.0000)	0.0014 (0.0000)	0.0035 (0.0000)
	30	0.0044 (0.0000)	0.0020 (0.0000)	0.0016 (0.0000)	0.0024 (0.0000)
	50	0.0038 (0.0000)	0.0023 (0.0000)	0.0018 (0.0000)	0.0015 (0.0000)
	200	0.0031 (0.0000)	0.0027 (0.0000)	0.0021 (0.0000)	0.0004 (0.0000)
	1000	0.0029 (0.0000)	0.0029 (0.0000)	0.0022 (0.0000)	0.0001 (0.0000)

Table 7: Mean [95% prediction interval] of each per-replication quantity, for the forced strategy, by item-pool size (k) and sample size (N). The interval is the empirical 2.5th to 97.5th percentile range across the 10,000 replications of a condition. It describes the spread across individual studies, not the precision of the mean, and unlike a Monte Carlo standard error it does not shrink as the number of replications grows.

k	N	3a. Observed gain	3c. True reliability change	2b. Bias vs. reliability
5	20	0.090 [-0.013, 0.360]	-0.001 [-0.107, 0.094]	0.024 [-0.267, 0.243]
	30	0.073 [-0.017, 0.289]	0.006 [-0.090, 0.097]	0.013 [-0.221, 0.199]
	50	0.058 [-0.019, 0.227]	0.011 [-0.074, 0.097]	0.006 [-0.173, 0.159]
	200	0.040 [-0.024, 0.166]	0.021 [-0.040, 0.099]	-0.008 [-0.103, 0.073]
	1000	0.036 [-0.025, 0.155]	0.023 [-0.028, 0.099]	-0.012 [-0.065, 0.027]
10	20	0.041 [0.002, 0.138]	0.006 [-0.025, 0.039]	-0.006 [-0.199, 0.111]
	30	0.034 [0.001, 0.110]	0.009 [-0.019, 0.040]	-0.007 [-0.155, 0.093]
	50	0.028 [0.000, 0.085]	0.011 [-0.015, 0.040]	-0.006 [-0.114, 0.073]
	200	0.022 [-0.001, 0.066]	0.014 [-0.006, 0.041]	-0.009 [-0.063, 0.034]
	1000	0.021 [-0.001, 0.062]	0.015 [-0.002, 0.043]	-0.009 [-0.036, 0.011]
15	20	0.024 [0.003, 0.075]	0.005 [-0.010, 0.021]	-0.010 [-0.156, 0.072]
	30	0.020 [0.003, 0.057]	0.006 [-0.007, 0.022]	-0.009 [-0.116, 0.061]
	50	0.017 [0.002, 0.046]	0.007 [-0.005, 0.023]	-0.008 [-0.088, 0.047]
	200	0.013 [0.002, 0.035]	0.009 [-0.001, 0.023]	-0.007 [-0.044, 0.022]
	1000	0.013 [0.001, 0.033]	0.009 [0.001, 0.023]	-0.007 [-0.025, 0.007]
20	20	0.016 [0.003, 0.047]	0.004 [-0.005, 0.013]	-0.010 [-0.122, 0.053]
	30	0.013 [0.003, 0.035]	0.004 [-0.004, 0.014]	-0.008 [-0.093, 0.044]
	50	0.011 [0.002, 0.028]	0.005 [-0.002, 0.014]	-0.007 [-0.066, 0.035]
	200	0.009 [0.002, 0.021]	0.006 [0.000, 0.014]	-0.006 [-0.034, 0.016]
	1000	0.008 [0.002, 0.020]	0.006 [0.001, 0.014]	-0.006 [-0.019, 0.005]
40	20	0.005 [0.001, 0.014]	0.001 [-0.001, 0.004]	-0.008 [-0.070, 0.025]
	30	0.004 [0.001, 0.010]	0.002 [-0.001, 0.004]	-0.006 [-0.051, 0.021]
	50	0.004 [0.001, 0.008]	0.002 [-0.000, 0.004]	-0.005 [-0.036, 0.017]
	200	0.003 [0.001, 0.006]	0.002 [0.001, 0.004]	-0.004 [-0.018, 0.008]
	1000	0.003 [0.001, 0.006]	0.002 [0.001, 0.004]	-0.003 [-0.010, 0.002]

3.4. What a single study can encounter

Every quantity above is a long-run average over 10,000 replications. The prediction intervals in Table 7 and Figure 5 describe the spread instead: the range of outcomes a single study, run once under this rule, can encounter.

The intervals narrow sharply as either design factor grows. For the bias in the reported α against true reliability, the interval spans 0.510 at $k = 5$, $N = 20$ ([-0.267, 0.243]) and only 0.012 at $k = 40$, $N = 1000$ ([-0.010, 0.002]). The true reliability change narrows correspondingly, from 0.200 to 0.003 wide.

The gap between these intervals and the corresponding Monte Carlo standard errors is the substantive point of this aim. At $k = 5$, $N = 20$, the mean bias against true reliability is 0.024 with a Monte Carlo standard error of 0.0013, so the long-run average is pinned down to within a thousandth. Yet the individual studies making up that average span [-0.267, 0.243], a range roughly 404 times wider than the standard error.

That ratio is not a finding about item dropping, it is close to an identity. For any approximately normal quantity the interval width is about $3.92S_d$ while the standard error is $S_d/\sqrt{K}$, so their ratio is about $3.92\sqrt{K} \approx 392$ at $K = 10{,}000$ regardless of the condition, and indeed it varies only between 345 and 424 across the entire design. Running more replications therefore raises this ratio rather than lowering it. The precision of the reported mean and the spread that an individual study faces are not merely different in size, they move in opposite directions as a simulation is scaled up. A researcher reporting an α after dropping an item from a five-item scale measured on twenty participants is drawing from the wide distribution, not from a neighbourhood of the mean, and no amount of additional simulation narrows what they face.

3.5. Post hoc analysis: can a stricter decision rule fix the decision?

The conditional strategy's gate, any apparent improvement at all, is the most permissive rule available, so an obvious practical question is whether a stricter gate rescues the practice: drop only if the apparent gain exceeds some threshold t. Because the simulation cache records the full-pool and dropped-pool sample alphas and both population targets for every replication, any such rule is computable from the same 250,000 replications without new simulation, with $t = 0$ reproducing the conditional strategy exactly. This analysis was not part of the planned design and should be read as post hoc. Table 8 reports, for $t \in \{0, 0.01, 0.02, 0.05\}$, how often each rule fires and the probability that firing harmed true reliability.

Table 8: Post hoc thresholded decision rules: drop only if the apparent in-sample gain exceeds t. For each rule and condition, the percentage of replications in which the rule fires, and the percentage of those firings that reduced true reliability (harm given fire). t = 0 is the conditional strategy. All values are percentages; the binomial Monte Carlo standard error is at most 0.5 percentage points for fire rates, and for harm rates it is computed on the realised number of firings. An em-dash marks cells in which the rule never fired, where harm given fire is undefined.

		Fires (% of replications)				Harm given fire (%)			
k	N	t = 0	t = .01	t = .02	t = .05	t = 0	t = .01	t = .02	t = .05
5	20	91.0	84.1	76.5	55.7	46.7	44.8	42.9	38.2
	30	88.7	80.9	72.9	50.5	39.4	36.7	34.2	28.6
	50	85.6	76.4	67.1	43.7	33.1	29.3	25.7	18.2
	200	80.1	69.7	59.4	33.6	16.0	11.2	7.0	1.8
	1000	77.9	66.9	56.0	30.0	7.2	2.2	0.6	0.0
10	20	98.7	86.6	68.4	28.0	32.0	28.9	25.4	19.2
	30	98.4	84.0	62.9	20.6	26.9	22.4	18.0	11.3
	50	97.5	80.2	56.3	13.9	19.8	14.0	9.0	4.1
	200	96.4	73.9	46.1	7.4	7.6	2.0	0.3	0.0
	1000	96.1	72.5	43.5	5.6	2.5	0.0	0.0	0.0
15	20	99.8	78.6	46.6	8.3	25.0	20.4	16.1	13.7
	30	99.8	74.8	39.0	4.2	19.0	13.7	8.7	5.3
	50	99.7	69.0	29.6	1.6	12.8	6.7	3.3	0.6
	200	99.4	59.0	18.9	0.3	3.7	0.3	0.1	0.0
	1000	99.2	57.0	16.3	0.2	0.8	0.0	0.0	0.0
20	20	100.0	63.8	25.0	2.1	20.4	15.4	13.1	14.6
	30	100.0	56.1	17.0	0.5	15.2	8.5	5.4	2.0
	50	99.9	47.6	9.6	0.1	8.9	3.1	1.6	0.0
	200	99.9	34.6	3.4	0.0	1.8	0.0	0.0	—
	1000	99.9	31.7	2.6	0.0	0.3	0.0	0.0	—
40	20	100.0	7.3	0.4	0.0	12.0	10.0	17.1	0.0
	30	100.0	2.7	0.1	0.0	7.0	4.0	0.0	—
	50	100.0	0.7	0.0	0.0	3.2	0.0	—	—
	200	100.0	0.1	0.0	0.0	0.1	0.0	—	—
	1000	100.0	0.0	0.0	0.0	0.0	—	—	—

The pattern splits cleanly on sample size. At $N \geq 200$, thresholds work: requiring an apparent gain of at least .02 keeps the harm-given-fire probability at or below 7.0% at $N = 200$ wherever the rule fires at all (7.0% at $k = 5$, 0.3% at $k = 10$), and near zero at $N = 1000$. At the longest scales the .02 gate simply never fires, because a single item out of 40 cannot move α by that much; abstention is the correct behaviour there, since those are also the cells where dropping has the least to offer. At small samples, thresholds fail: at $k = 5$, $N = 20$, even demanding an apparent gain of .05, which more than halves how often the rule fires, still leaves a 38.2% chance that the drop harms true reliability, and at $N = 50$ the strictest gate examined still carries 18.2%. The reason is the one documented under Aim 4: at small N the apparent gain is dominated by selection over noise, so conditioning on a *larger* apparent gain selects for more extreme noise nearly as often as for genuinely worse items, and no threshold on an in-sample quantity can separate the two.

As a rule of thumb, an alpha-if-item-removed recommendation should not be acted on at all in small samples ($N \leq 50$), whatever the apparent improvement; in samples of several hundred or more, acting only when the apparent gain exceeds .02 keeps the risk of harming the scale below roughly one in fourteen even for five-item scales, and negligible for longer ones.

4. Discussion

Four things follow from these results. First, α has two independent reasons to be untrustworthy before any item is dropped at all: ordinary small-sample noise, which fades with N (from -0.042 at $N = 20$ to -0.0008 at $N = 1000$ for $k = 5$), and a structural ceiling on what α can measure relative to true reliability, which does not fade with N at all, only with k (from -0.024 at $k = 5$ to -0.004 at $k = 40$). Second, introducing the selection step adds a third source of distortion that interacts with the first two and flips the overall sign of the bias depending on where in the design a study sits. Third, the bias in the number is almost beside the point: the decision itself is frequently wrong in the sense that matters, degrading true reliability. Fourth, none of these averages is a safe guide to what any one real study will observe.

4.1. What we expected, and what we found

Because this simulation was not preregistered, the expectations recorded in the Method are priors rather than tests, and we set them against the results explicitly so that the reader can see which parts of the account were anticipated and which were not.

Three expectations held straightforwardly. The full-pool calibration bias was small, negative, and shrank toward zero with N at every item-pool size. The structural gap was negative in every one of the 25 conditions. And the prediction intervals were wide relative to the means wherever selection had room to operate, narrowing as either design factor grew.

One expectation was clearly wrong, and its being wrong is informative. We expected selection optimism to be larger at larger item-pool sizes, reasoning that more candidate items gives a larger expected maximum over noise. The opposite holds: the bias against the population α of the retained items is largest at $k = 5$ (0.0435 at $N = 20$) and is already slightly negative at $k \geq 15$ with small samples. Our reasoning counted the number of candidates and ignored their individual leverage. With more items each contributes proportionally less to α, so the maximum is taken over a larger set of individually smaller perturbations, and the second effect dominates. It is also swamped, at longer pools, by the small-sample downward calibration bias pulling in the opposite direction. The practical implication runs opposite to the intuition: long scales are not more exploitable by this rule, they are less so.

A second expectation was right in direction but badly wrong in emphasis. We expected the reliability-scale bias to be negative at large N and possibly positive at small N, and treated the location of the sign change as the open question. It is positive in only 3 of 25 conditions, all at the shortest pool size, so the sign change is real but confined to a corner of the design. Framing the study around it, as an “alpha inflation” account would, would have been a mistake. Across most of the grid the reported number is only slightly wrong, and had we stopped at the estimands we thought would carry the paper, we would have concluded that item dropping is a minor problem. The harm rate, for which we

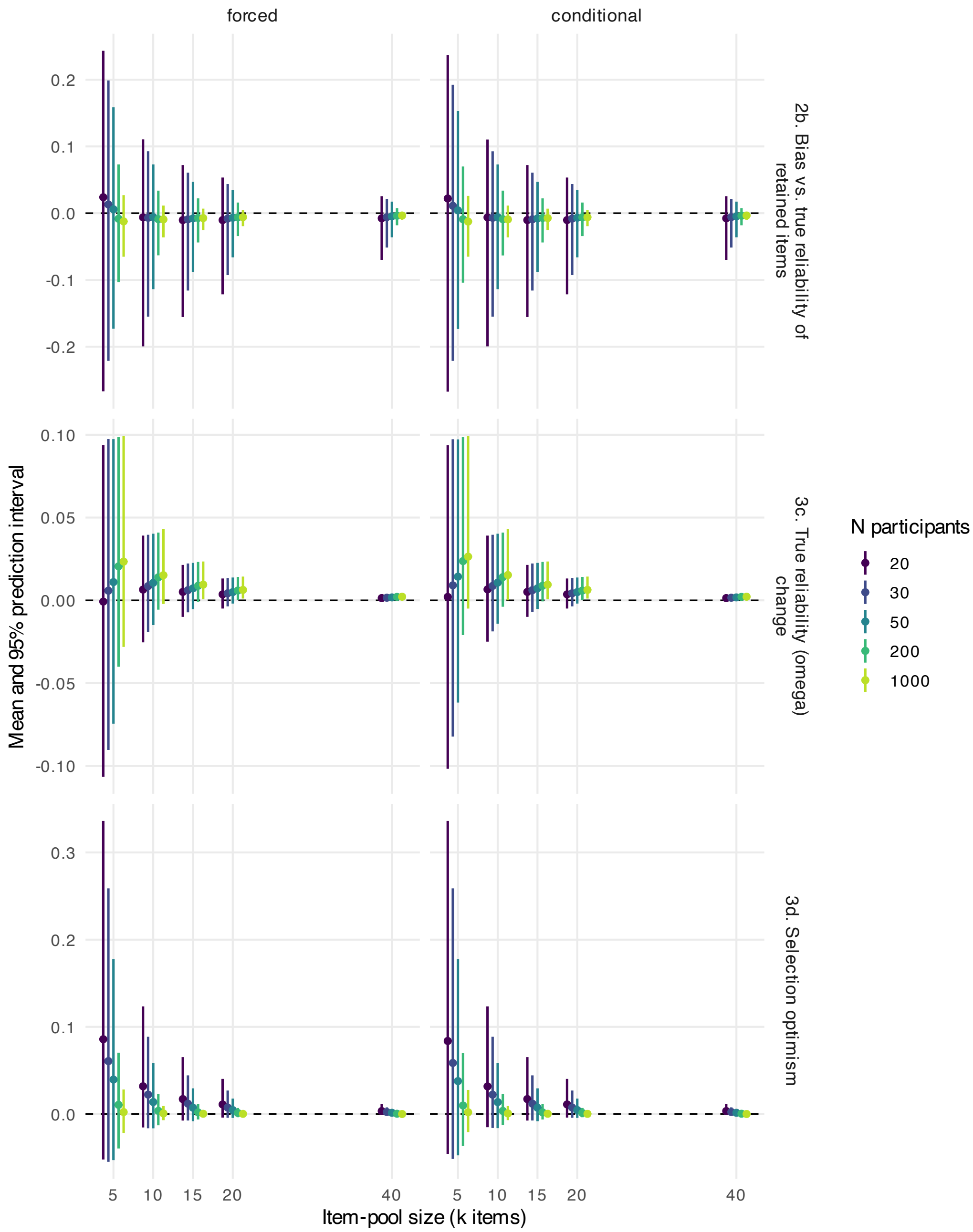

Figure 5: Mean and 95% prediction interval for the bias in the reported alpha against true reliability (top), the true reliability change (middle), and the selection optimism (bottom), for the forced (left) and conditional (right) strategies, by item-pool size and sample size. Points are means, bars span the empirical 2.5th to 97.5th percentiles across replications. Bars are dodged horizontally within each item-pool size so the five sample sizes are distinguishable. Note the differing y-axis scales between rows.

deliberately made no directional prediction because we could not anticipate the balance between genuinely bad items and pure length penalties, is what shows otherwise: up to 49.6% of replications made worse. The finding we did not predict is the one that matters, and it is measured by an estimand we included precisely because we did not know what it would show.

Finally, Pr(drop) rose with item-pool size as expected, but its behaviour in N was the informative part and we had not thought it through in advance. It falls as N grows at short pool sizes, from 91.0% to 77.9%, because noise-driven drops die out with sampling precision and only genuinely harmful items are left to remove. The rule therefore fires most often exactly where its firing means least.

4.2. Reconciling with the simulation literature

Kopalle & Lehmann (1997) established that item selection based on the same sample subsequently used to report the resulting α capitalises on chance. The present work reproduces that finding where the item pool and sample are both small, and qualifies it elsewhere. Measured against the population α of the retained items, the optimism is unambiguous at $k = 5$ but has essentially vanished by $k = 15$, where it is outweighed by the opposing small-sample calibration bias. Measured against true reliability, which is the quantity researchers care about, the sign reverses across most of the design. The direction of bias in an α reported after item dropping is therefore not fixed, and an unqualified "alpha inflation" framing does not survive contact with a design that varies scale length and sample size over realistic ranges and evaluates against reliability rather than against α itself.

The finding that α and true reliability diverge even with no item selection at all is consistent with a separate, long-running critique of Cronbach's α. Sijtsma (2009) argued that α cannot in general equal the true reliability of a scale given its inter-item covariance structure, a point developed in the α/ω literature (McDonald, 1999; Revelle & Zinbarg, 2009; Zinbarg et al., 2005), which shows the two coefficients are equal only under the restrictive condition of essentially equal item loadings. Our structural-gap result is a direct simulation demonstration of that theoretical point under an empirically calibrated loading distribution, and shows its magnitude is not negligible relative to the biases usually discussed in the item-dropping literature. At the shortest scale examined it is 0.024, which exceeds the selection optimism observed anywhere in the design outside the $k = 5$ column.

4.3. Bias in the reported number versus harm to the scale

The most consequential finding is not that the reported number is wrong. Across most of the design it is only slightly wrong, with magnitudes mostly under 0.01. It is that the decision itself is frequently detrimental. In the worst condition, the forced rule reduced true reliability in 49.6% of replications while its mean apparent in-sample gain was 0.090. The conditional rule, which acts only on an apparent improvement, still did so in 46.7% of the replications in which it acted.

A modest average bias can coexist with a near-coin-flip probability of a harmful outcome, because the bias measure averages over both helpful and harmful replications while the harm rate isolates how often the harmful case occurs. Nothing about the reported α itself distinguishes the two scenarios.

This bears on how the downstream result should be read. Stefan & Schönbrodt (2023) show that item dropping on α inflates the false positive rate of the subsequent hypothesis test to as high as 30%, and it would be natural to attribute that to the reported reliability being overstated. These results suggest that is largely not the mechanism: across most of the design the reported α is close to correct, and in the majority of conditions it is too low rather than too high. What the practice supplies instead is a decision point at which the analyst may retain either of two scales, taken after seeing the data, in conditions where the two are close to indistinguishable on the evidence available. The inflation is better understood as an experimenter degree of freedom than as a mis-estimated reliability, which matters because the two call for different remedies: reporting ω instead of α would address the second and not the first. This is also why the average is the wrong summary to lead with for practice: a mean Δ_ρ of -0.0007 at $k = 5$, $N = 20$ reads as "no effect" but is the average of a distribution in which roughly half of studies are made worse and roughly half are made better.

4.4. Mean bias understates practical risk

A mean bias near zero is compatible either with every study landing close to zero or with studies scattered across a wide range that happens to average out there. Table 7 shows the design is unambiguously the second case for short scales: the $k = 5$, $N = 20$ cell alone spans [-0.267, 0.243]. Additional Monte Carlo replications would narrow the standard error on the mean further but would not narrow this interval by anything, since it is a property of the distribution rather than of how many times we sampled from it. Reporting means with Monte Carlo standard errors alone, which is common practice in simulation studies, would understate what item dropping can do in a single study by a factor of roughly 392 here, and by a larger factor still in any simulation run with more replications.

4.5. Practical implications

In short scales and small samples, precisely where the reported number is most likely to be optimistically biased, the underlying decision is most likely to be actively wrong. Previous simulation findings have shown that unbiased estimation of α at small N is possible when the largest eigenvalue of the item covariance matrix is large (Yurdugül, 2008), indicating that bias in α depends on how strongly unidimensional the items are as well as on sample size. Every pool generated here is unidimensional by construction (see Limitations), so this moderator could not vary within our design; scales with weaker general factors than ours would likely fare worse, not better.

The key distinction for measurement practice is between what is knowable before a study is run and what is knowable after. Scale length and sample size are properties of the study design. Everything this simulation quantifies as risky, that is, the optimistic bias, the harm rate, and the width of the prediction interval, is a function of those two design choices and of nothing about the data subsequently collected, holding the loading distribution fixed at the trait-scale-typical one simulated here. The risk can therefore be known in advance for scales resembling that corpus: a researcher could take a proposed k and N, find the corresponding row of Table 5, and know before collecting data roughly how much to trust a future item-dropping decision from that study. This is an unusual property for a bias to have. Many statistical biases depend on unknown population parameters that the researcher cannot inspect in advance. This one depends only on design choices that are already written down in the preregistration or method section, together with the assumption that the scale in hand is exchangeable with the trait-scale pool that calibrated the simulation; a scale with unusually strong and uniform loadings faces lower risk at the same k and N, and a weaker one higher (see the clustering limitation below).

The post hoc decision-rule analysis sharpens this advice. Where a large sample is available, a simple threshold works: act on an alpha-if-item-removed recommendation only when the apparent gain exceeds .02, which at $N \geq 200$ kept the probability of harming the scale at or below 7.0% wherever the rule recommended dropping at all. Where the sample is small, no threshold examined made the decision safe, so the recommendation is to not act at all and to treat the table as uninformative.

This gives a partial answer to the question Hussey et al. (2025) raise, of how a researcher can tell a genuinely poor item from an artefact of their own sample without a second sample to check against. The answer these results support is that they cannot tell from the analysis, but they can bound how often they would be wrong from the design. At $k = 40$ with $N = 1000$ that bound is negligible. At $k = 5$ with $N = 20$ it is close to a coin flip.

What cannot be known in advance is anything from inside the alpha-if-item-removed table itself. The table that `psych::alpha()`, or any equivalent, produces looks exactly the same whether the item it nominates is genuinely the weakest link in the scale or is the item that happened to draw an unfavourable sample of participants this time. It carries no confidence interval and no indication of ranking instability. Two researchers who drew two different samples of the same size from the same true scale would each see a complete, definitive-looking table, possibly nominating different items for removal, and each would have exactly the same apparent justification for acting on it, even though at most one of them has observed real signal. The safeguard has to come from knowing where k and N sit in a risk table such as Table 5, because nothing internal to the analysis will indicate the risk level.

4.6. Limitations

The main limitation is that both strategies consider dropping at most one item, in one step. Neither iterates nor targets a final scale length, so multi-item dropping is not simulated but is instead noted as a future direction below.

The population targets are realised rather than idealised subset targets: they are the true properties of whichever items a given replication's procedure retained, not of the best possible subset. This is the right target for the applied question, but it means the targets vary by strategy and by replication rather than being a single fixed value per condition, a deliberate departure from the classical estimation setup that we flag so it is not mistaken for an oversight. Relatedly, only sample α is evaluated as the reported estimator. Sample ω is never estimated from data anywhere in this design, it is used only as a population-level target.

The 95% prediction intervals conflate two sources of variation, a fresh item pool and a fresh sample in each replication, so they support a field-level claim about studies of trait scales in general rather than a claim about resampling from one fixed scale. Separating the two would require a nested design with several samples per drawn loading vector. This is the same structure that the item-choice replicability question below requires. As such, the two are deferred to a single follow-up study.

Finally, data-generating processes vary considerably between simulation studies, and identifying where ours limits our conclusions is worthwhile despite its being empirically grounded rather than speculative. The corpus used to fit the loading distribution consists of individual-differences and personality measures, which may differ considerably from the scales on which item-dropping decisions are made in clinical psychology or in contexts where psychometric properties carry more consequential weight.

Two idealisations of the generated data itself also bound the generality of the results. First, items are continuous and multivariate normal, whereas almost every α in the motivating literature is computed on ordinal Likert responses. Categorical items attenuate inter-item covariances, α computed on them behaves differently at small samples, and item dropping could interact with asymmetric response thresholds in ways a continuous DGP cannot exhibit. Second, every generated pool is exactly unidimensional: the single-factor model that α presupposes, short of tau-equivalence, is true by construction, so the only kind of genuinely weak item that can exist here is one with a low or negative loading. In practice, alpha-if-item-removed nominations are often driven by multidimensionality, cross-loadings, or correlated residuals, cases in which the nominated item is a signal of model misspecification rather than of noise, and in which removing it changes what the scale measures rather than merely how precisely. Our conclusion that the table cannot distinguish a weak item from a sampling artefact therefore describes the *best case* for the practice, the one in which the model the table implicitly assumes is correct. Simulation designs that build in realistic indicator distributions and deliberate misspecifications, in the manner of Greene et al. (2019), who simulated binary and skewed indicators under a range of structural misspecifications precisely because fit-based model comparisons behave differently there, would test whether the harm rates reported here are under- or over-estimates for real data.

More consequentially, the empirical loading distribution is a marginal, independent-item fit: each item's loading is drawn i.i.d. from the one pooled distribution, ignoring which items belong to which scale. Real loadings cluster within scales. Some scales are uniformly well constructed, while weak and negative loadings concentrate within particular, often multidimensional, scales. Under the i.i.d. fit that clustering is erased: genuinely weak items are sprinkled uniformly at random across generated pools, every sufficiently long pool contains some droppable item, and no pool is systematically better constructed than any other. The fit reproduces the marginal distribution of loadings faithfully, but it understates how much pools differ from one another, because clustering concentrates item quality within scales and the i.i.d. draw spreads it evenly. This matters for the estimands most central to this article: the harm rate and the probability of dropping both turn on whether the nominated item is a stable property of the scale or an accident of the sample, and a DGP that understates between-scale heterogeneity misstates the balance between those two ingredients, and with it how the marginal harm rate is composed.

The same flattening bears directly on a second limitation, noted above: this simulation says nothing about whether an item-dropping recommendation would *replicate*. Each replication draws one sample per pool, so the design can estimate how often a drop harms true reliability, but not whether the same item would be nominated again in a new sample from the same scale, nor whether an apparent in-sample gain would survive out of sample. These two limitations are not independent, because a recommendation replicates exactly when it is driven by the stable, scale-level component of the loadings rather than by sampling noise, and that is precisely the component the marginal fit flattens away. Both therefore point at the same future simulation: one that parametrises the loading distribution hierarchically, so that scales genuinely differ from one another, and that splits each pool's data into training and testing samples, nominating the item in the training sample and assessing the decision's out-of-sample performance in the testing sample: whether the same item is nominated again, whether the reported α holds up, and whether the apparent gain in reliability is realised. The Future directions below sketch this design.

4.7. Future directions

The replicability question raised in the limitations could be investigated with a same-item agreement measure benchmarked against a uniform chance floor rather than against Cohen's kappa, and with a paired training/testing design that distinguishes whether the same item gets nominated twice from whether a fixed item's apparent gain survives out of sample. Because that design draws several samples from each item pool, it would also decompose the harm rate reported here. A marginal harm rate of close to one half is compatible with two quite different worlds: one in which every short scale is a coin flip, and one which mixes scales with no genuinely weak item, where dropping is near-certain to harm, with scales containing one, where it is near-certain to help. These carry opposite practical advice, the present design cannot distinguish them, and the nested follow-up can. Combining that analysis with the bias and harm-rate results here would complete the picture this pair of simulations is building toward.

A second extension would make the generated data more realistic on the two dimensions identified in the Limitations: ordinal indicators, generated by thresholding the latent responses at empirically calibrated cut points, and deliberately misspecified structures, such as minor secondary factors, cross-loadings, or correlated residuals, following the approach of Greene et al. (2019) of simulating realistic indicator distributions under a range of structural misspecifications. The misspecified case is the more informative one, because it is the only setting in which an alpha-if-item-removed nomination can be *right for the wrong reason*, flagging an item that genuinely does not belong while the analyst's unidimensional reading of the scale is itself in error.

A third extension would evaluate sample ω as an alternative reporting statistic, since the present study uses ω only as the population target against which sample α is judged rather than as a competing sample-based estimator. A fourth would add an external criterion, which would let the validity consequences of dropping be estimated directly rather than inferred from the reliability change. Finally, whether the specific practices simulated here reflect how item dropping actually occurs in the published literature remains an open empirical question.

4.8. Conclusion

Item dropping to improve Cronbach's α introduces bias into psychological measurement, and in more than one direction at once. In short scales and small samples it reduces true reliability in close to half of simulated studies, including when the rule acts only on an apparent in-sample improvement. In longer scales and larger samples it produces the opposite distortion, making the scale's reliability appear slightly lower than it truly is. Neither distortion is visible from inside the analysis that produces it. What is visible in advance is the study design, and researchers planning short scales with small samples should treat an alpha-if-item-removed recommendation from such a study as close to uninformative.

Author notes

CRediT authorship contribution statement

ZF: Writing – original draft. IH: Conceptualization, Methodology, Software, Formal analysis, Data curation, Visualization, Writing – review & editing.

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Conflict of interest

The authors declare that they have no conflicts of interest.

Data and code availability

All simulation code and results, including the full ADEMP specification, the validation of the reliability functions, per-condition tables for every estimand, and Monte Carlo standard errors for every estimate, are available at https://github.com/ianhussey/replicability-of-item-dropping.

References

Anvari, F., Alsalti, T., Oehler, L. A., Hussey, I., Elson, M., & Arslan, R. C. (2025). Defragmenting psychology. *Nature Human Behaviour*, 1–4. https://doi.org/10.1038/s41562-025-02138-0

Anvari, F., Alsalti, T., Oehler, L. A., Marion, Z., Hussey, I., Elson, M., & Arslan, R. C. (2025). A Fragmented Field: Construct and Measure Proliferation in Psychology. *Advances in Methods and Practices in Psychological Science*, *8*(3), 25152459251360642. https://doi.org/10.1177/25152459251360642

Clark, L. A., & Watson, D. (1995). Constructing validity: Basic issues in objective scale development. *Psychological Assessment*, *7*(3), 309–319. https://doi.org/10.1037/1040-3590.7.3.309

Clark, L. A., & Watson, D. (2019). Constructing validity: New developments in creating objective measuring instruments. *Psychological Assessment*, *31*(12), 1412–1427. https://doi.org/10.1037/pas0000626

Cortina, J. M. (1993). What is coefficient alpha? An examination of theory and applications. *Journal of Applied Psychology*, *78*(1), 98–104. https://doi.org/10.1037/0021-9010.78.1.98

Cronbach, L. J. (1951). Coefficient alpha and the internal structure of tests. *Psychometrika*, *16*(3), 297–334.

Elson, M., Hussey, I., Alsalti, T., & Arslan, R. C. (2023). Psychological measures aren't toothbrushes. *Communications Psychology*, *1*(1), 25. https://doi.org/10.1038/s44271-023-00026-9

Flake, J. K., & Fried, E. I. (2020). Measurement Schmeasurement: Questionable measurement practices and how to avoid them. *Advances in Methods and Practices in Psychological Science*, *3*(4), 456–465. https://doi.org/10.1177/2515245920952393

Flake, J. K., Pek, J., & Hehman, E. (2017). Construct Validation in Social and Personality Research: Current Practice and Recommendations. *Social Psychological and Personality Science*, *8*(4), 370–378. https://doi.org/10.1177/1948550617693063

Greene, A. L., Eaton, N. R., Li, K., Forbes, M. K., Krueger, R. F., Markon, K. E., Waldman, I. D., Cicero, D. C., Conway, C. C., Docherty, A. R., Fried, E. I., Ivanova, M. Y., Jonas, K. G., Latzman, R. D., Patrick, C. J., Reininghaus, U., Tackett, J. L., Wright, A. G. C., & Kotov, R. (2019). Are fit indices used to test psychopathology structure biased? A simulation study. *Journal of Abnormal Psychology*, *128*(7), 740–764. https://doi.org/10.1037/abn0000434

Heggestad, E. D., Scheaf, D. J., Banks, G. C., Monroe Hausfeld, M., Tonidandel, S., & Williams, E. B. (2019). Scale adaptation in organizational science research: A review and best-practice recommendations. *Journal of Management*, *45*(6), 2596–2627. https://doi.org/10.1177/0149206319850280

Hussey, I. (2026). *Granary: A very large item-level open dataset of psychological traits and attitudes*. https://doi.org/10.5281/zenodo.21620120

Hussey, I., Alsalti, T., Bosco, F., Elson, M., & Arslan, R. (2025). An Aberrant Abundance of Cronbach's Alpha Values at .70. *Advances in Methods and Practices in Psychological Science*, *8*(1), 25152459241287123. https://doi.org/10.1177/25152459241287123

Hussey, I., & Hughes, S. (2020). Hidden invalidity among 15 commonly used measures in social and personality psychology. *Advances in Methods and Practices in Psychological Science*, *3*(2), 166–184. https://doi.org/10.1177/2515245919882903

Jebb, A. T., Ng, V., & Tay, L. (2021). A review of key Likert scale development advances: 1995–2019. *Frontiers in Psychology*, *12*. https://doi.org/10.3389/fpsyg.2021.637547

Joshi, M., & Pustejovsky, J. E. (2022). *Simhelpers: Helper functions for simulation studies*. https://CRAN.R-project.org/package=simhelpers

Kopalle, P. K., & Lehmann, D. R. (1997). Alpha inflation? The impact of eliminating scale items on Cronbach's alpha. *Organizational Behavior and Human Decision Processes*, *70*(3), 189–197. https://doi.org/10.1006/obhd.1997.2702

Lüdecke, D., Ben-Shachar, M. S., Patil, I., Waggoner, P., & Makowski, D. (2021). Performance: An R package for assessment, comparison and testing of statistical models. *Journal of Open Source Software*, *6*(60), 3139. https://doi.org/10.21105/joss.03139

McDonald, R. P. (1999). *Test theory: A unified treatment*. Lawrence Erlbaum.

Morris, T. P., White, I. R., & Crowther, M. J. (2019). Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, *38*(11), 2074–2102. https://doi.org/10.1002/sim.8086

Revelle, W. (2024). *Psych: Procedures for psychological, psychometric, and personality research*. https://CRAN.R-project.org/package=psych

Revelle, W., & Zinbarg, R. E. (2009). Coefficients alpha, beta, omega, and the glb: Comments on Sijtsma. *Psychometrika*, *74*(1), 145–154. https://doi.org/10.1007/s11336-

008-9102-z
Rodriguez, A., Reise, S. P., & Haviland, M. G. (2016). Applying bifactor statistical indices in the evaluation of psychological measures. *Journal of Personality Assessment*, *98*(3), 223–237. https://doi.org/10.1080/00223891.2015.1089249
Rosseel, Y. (2012). Lavaan: An R package for structural equation modeling. *Journal of Statistical Software*, *48*(2), 1–36. https://doi.org/10.18637/jss.v048.i02
Siepe, B. S., Bartoš, F., Morris, T. P., Boulesteix, A.-L., Heck, D. W., & Pawel, S. (2024). Simulation studies for methodological research in psychology: A standardized template for planning, preregistration, and reporting. *Psychological Methods*. https://doi.org/10.1037/met0000695
Sijtsma, K. (2009). On the use, the misuse, and the very limited usefulness of Cronbach's alpha. *Psychometrika*, *74*(1), 107–120. https://doi.org/10.1007/s11336-008-9101-0
Stefan, A. M., & Schönbrodt, F. D. (2023). Big little lies: A compendium and simulation of p-hacking strategies. *Royal Society Open Science*, *10*(2), 220346. https://doi.org/10.1098/rsos.220346
Yurdugül, H. (2008). Minimum sample size for Cronbach's coefficient alpha: A Monte-Carlo study. *Hacettepe Üniversitesi Eğitim Fakültesi Dergisi*, *35*(35), 1–9.
Zinbarg, R. E., Revelle, W., Yovel, I., & Li, W. (2005). Cronbach's alpha, Revelle's beta, and McDonald's omega_h: Their relations with each other and two alternative conceptualizations of reliability. *Psychometrika*, *70*(1), 123–133. https://doi.org/10.1007/s11336-003-0974-7